DP-800

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https://www.youtube.com/watch?v=76WbJe9KNdU&list=PLMi6KgK4_mk3iq1z3TaFibTO9rSU_vKkZ&index=3

https://www.youtube.com/watch?v=k9NaAjZcchA

https://www.youtube.com/watch?v=9MnHXM43CLw&list=PLyABYqulvUwa_oIuo_KRKZZK03AsX1Wpv&pp=0gcJCe4COCosWNin

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https://www.koreadumps.com/DP-800-practice-test.html#

QnA

https://www.youtube.com/watch?v=E3Zld2-vGUo

Q- You are designing an AI-enabled application that uses an Azure SQL Database to store vector embeddings for semantic search. You need to ensure the system can accurately find similar documents based on user queries. Which two techniques should you implement? Each correct answer presents part of the solution.

A. Calculate the cosine distance between the search vector and stored vectors.

B. Create a standard relational table to store the vectors as simple text arrays.

C. Use a built-in full text search catalog to index the floating point numbers.

D.Store the embeddings in a column optimized for vector data.

E. Implement a traditional primary key constraint to cluster the vectors.

Q- You want to automatically generate vector embeddings for new product descriptions directly within your Azure SQL Database immediately after the data is inserted. Which approach minimizes external application dependencies and external compute?

A. Trigger a serverless function that reads the database and calls the language model.

B. Export the daily changes to a storage account for a separate machine learning pipeline.

C. Deploy a containerized background service that polls the product table for new entries.

D. Use native database system procedures to invoke the external artificial intelligence endpoint.

Q- Your development team is configuring an automated pipeline to deploy schema updates to an Azure SQL Database. The security team requires that the pipeline authenticates without using static passwords. What is the most appropriate authentication method to use for this deployment?

A. An active directory service principal with federated identity credentials.

B. A standard database user account with administrative privileges.

C. A long-lived database connection string stored as a pipeline secret.

D. An anonymous endpoint configured with a specialized firewall rule.

Q- You are building a generative AI solution that retrieves customer profiles from an Azure SQL Database. You need to ensure that the AI model cannot access the actual credit card numbers or personal phone numbers, but the application can still process the regional demographics of those users. Which two security features should you implement? Each correct answer presents part of the solution.

A. Row level security to hide the entire customer record.

B. Column level permissions to restrict access to the financial data.

C. Transparent data encryption to protect the data at rest.

D. A secure enclave to process the demographic queries in memory.

E. Dynamic data masking to obscure the sensitive contact fields.

Q- An external data provider sends real-time telemetry data formatted as complex JSON documents. You need to extract specific nested values from these documents to filter incoming records before permanently storing them in your relational tables. Which strategy provides the best performance for reading these specific values during the ingestion process?

A. Store the entire document in a text column and parse it in the application layer.

B. Convert the document to an external file format and query it using external tables.

C. Use built-in database functions to extract the required scalar values from the document.

D. Shred the entire document into multiple relational tables before applying any filters.

Link: https://www.youtube.com/watch?v=oiV3hAm_Xgk

Q- Your IT company is designing a generative AI solution that uses a Microsoft SQL Server 2024 database named SalesDB as its data source. The solution must:
Generate responses grounded in the latest transactional and reference data stored in SalesDB
Avoid retraining or fine-tuning the language model when the data changes
Provide citations or references to the source data used in responses
Which scenario is the best use case for implementing a Retrieval Augmented Generation (RAG) pattern?

A. Create summaries from unstructured customer feedback text.

B. Build a custom language model trained on historical SalesDB records.

C. Respond to employee queries using company-specific knowledge stored in SalesDB.

D. Generate promotional slogans based on sentiment analysis insights.

Answer -C

Q- Your IT company manages an Azure SQL database with a table named dbo.Inventory. The table contains three columns: VectorData, ProductType, and Cost. The VectorData column is defined as VECTOR(1536) and supports semantic search using AI_GENERATE_EMBEDDINGS and VECTOR_SEARCH, with filters applied on ProductType and Cost. You plan to switch the embedding model from text-embedding-ada-002 to text-embedding-3-small. Existing rows already contain embeddings in the VectorData column. You must ensure applications continue to use VECTOR SEARCH without runtime errors after the change. What should you do first?

A. Recompute embeddings for all existing records in dbo.Inventory.

B. Apply normalization to vector values before saving new embeddings.

C. Alter the VectorData column type to nvarchar(max).

D. Build a vector index on the VectorData column in dbo.Inventory.

Answer-A

Q- Your IT company uses a GitHub Actions workflow to build and deploy an Azure SQL database The schema is maintained in a GitHub repository as an SDK-style SQL database project. After a code review, you are asked to generate a report that shows whether the production schema has diverged from the model in source control.
Which action should be added to the pipeline?
A. Run SqlPackage.exe /Action: DriftReport to detect schema differences.

B. Run SqlPackage.exe /Action:DeployReport to preview deployment changes.

C. Run SqlPackage.exe /Action:Extract to capture the current database schema.

D. Run SqlPackage.exe /Action:Script to generate a deployment script.

Answer-A

Q- Your IT company runs an Azure SQL database that stores order records. A reporting query that aggregates periodic revenue per customer is executed frequently. You need to improve query performance and reduce retrieval time for the calculated values. The solution must not alter any underlying table structure. What should you configure?

A. Define a view with ORDER BY (no TOP) and place a unique clustered index on it.

B. Build a view without WITH SCHEMABINDING and add a nonclustered index.

C. Create a view using GROUP BY and attach a unique clustered index.

D. Implement a view with WITH SCHEMABINDING, include COUNT_BIG()*, and apply a unique clustered index.

Answer-D

Q- Your IT company manages an Azure SQL database that contains a column named Remarks. During a security audit, it was discovered that Remarks contains sensitive information. You must ensure that the data is protected so that neither the stored values nor the query inputs reveal the actual content. The solution must also prevent users from inferring relationships or repetitions in the encrypted output. Which approach should you use?

A. Use Always Encrypted with secure enclaves.

B. Use Always Encrypted with randomized encryption.

C. Apply row-level security (RLS).

D. Use Always Encrypted with deterministic encryption.

Answer-B

Q- You are working in an IT services company where an Azure SQL database supports the OLTP workload of a customer order-processing system. During a 15-minute incident, you query dynamic management views and observe:
Session 105 is idle but shows open transaction count = 1.
Several other sessions list blocking session_id = 105 in sys.dm_exec requests.
The input buffer for session 105 only shows BEGIN TRANSACTION UPDATE Sales.Orders.
Employees report that updates to Sales.Orders intermittently time out until you manually terminate session 105. What is the likely cause of the blocking?
A. A reporting query with a long-running SELECT is holding locks on the table.

B. Session 105 encountered a deadlock situation.

C. A transaction was started but never committed or rolled back.

D. A lock escalation occurred on the Sales.Orders table.

Answer-C

Q- Your IT company operates a customer-facing web application backed by an Azure SQL database. A stored procedure named dbo.FetchClientOrders is executed thousands of times each day by the API. After a recent deployment that modified indexes and statistics, users complain that the API endpoint has slowed down. In Query Store, you notice two persisted execution plans for the same query. The newer plan shows much higher average duration and CPU usage compared to the older plan. You need to quickly restore the previous performance without altering the API code. Which Transact-SQL command should you execute?
A. EXEC sys.sp_query_store_set_hints - Apply query hints through Query Store.

B. DBCC FREEPROCCACHE - Clear cached execution plans from memory.

C. EXEC sp_query_store_force_plan - Force Query Store to use a specific execution plan.

D. ALTER DATABASE - Change database-level configuration settings.

Answer-C

Q- Your IT company manages an internal support knowledge base in Azure SQL Database. The table dbo.SupportDocs stores technical articles with columns:

- ArticleID (int)
- Heading (nvarchar(200))
- Content (nvarchar(max))
- UpdatedOn (datetime2) Another table, dbo.SupportDocs Embeddings, stores embeddings for semantic search with columns
- ArticleID (int)
- SectionOrder (int)
- SectionText (nvarchar(max))
- VectorData (vector(1536))

Currently, embeddings are refreshed once each day during the night. Management wants embeddings to be updated whenever article content changes, without relying on nightly batch jobs. What should the solution include?
A. Fixed-size chunking.

B. Smaller embedding model.

C. Table triggers. >>Answer: C

Q- Your IT company hosts a knowledge repository in Azure SQL Database named SupportDB. The table dbo.KnowledgeArticles stores millions of technical articles with embeddings. These articles are updated frequently throughout the day. Engineers query embeddings using VECTOR_SEARCH. Users complain that semantic search results do not reflect the latest updates until the next day. You must ensure embeddings are refreshed whenever articles change, while keeping CPU usage on SupportDB low. Which embedding maintenance method should be implemented?

A. Use VECTOR_DISTANCE instead of VECTOR_SEARCH.

B. Enable change data capture (CDC) on dbo.KnowledgeArticles and process CDC changes with an Azure Functions app.

C. Run an hourly SQL job to regenerate embeddings for all rows in dbo.KnowledgeArticles.

D. Create a trigger on dbo.KnowledgeArticles that calls AI_GENERATE_EMBEDDINGS for each inserted or updated row.

Answer: B

Q- You are working in an IT company where an Azure SQL database named StaffRecordsDB stores employee details in the table dbo.Worker. A C# Azure Functions app already uses an HTTP-triggered function with SQL input binding to query dbo.Worker. Now, management wants a second function that should react to row changes in dbo.Worker and generate structured logs. The requirements are:
• The function must be triggered within seven seconds of a change.
• Each run should process no more than 120 changes. Which TWO database configurations should you apply?
A. Create an AFTER trigger on dbo.StaffRecords for DML operations.

B. Configure Sql_Trigger_MaxBatchSize = 120.

C. Enable change tracking on dbo.StaffRecords.

D. Set Sql_Trigger_PollingIntervalMs to 5000.

Answer:B,D

Link: https://www.youtube.com/watch?v=xLuVLEiaQMw

Q-You need to enable similarity search to provide the analysts with the ability to retrieve the most relevant health summary reports. The solution must minimize latency.
What should you include in the solution?
A. a computed column that manually compares vector values
B. a standard nonclustered index on the Embeddings (vector(1536)) column
C. a full-text index on the Embeddings (vector(1536)) column
D. a vector index on the Embeddings (vector(1536)) column
Answer: D

Q-You have an Azure SQL database that contains database-level Data Definition Language (DDL) triggers, including a trigger named ddl_Audit.
You need to prevent ddl_Audit from firing during the next deployment. The trigger object must remain in place.
Which Transact-SQL statement should you use?
A. DISABLE TRIGGER
B. ALTER TRIGGER
C. ALTER DATABASE
D. ALTER DATABASE AUDIT SPECIFICATION
E. ALTER SERVER AUDIT SPECIFICATION
Answer: A

Q-You need to recommend a solution that will resolve the ingestion pipeline failure issues.
Which two actions should you recommend? Each correct answer presents part of the solution.
A. Enable snapshot isolation on the database.
B. Use a trigger to automatically rewrite malformed JSON.
C. Add foreign key constraints on the table.
D. Create a unique index on a hash of the payload.
E. Add a check constraint that validates the JSON structure.
Answer: D,E

Q-You have a SQL database in Microsoft Fabric that contains a table named dbo.Orders. dbo.Orders has a clustered index, contains three years of data, and is partitioned by a column named OrderDate by month.
You need to remove all the rows for the oldest month. The solution must minimize the impact on other queries that
access the data in dbo.Orders.
Solution: Identify the partition scheme for the oldest month, and then run:
ALTER TABLE dbo.Orders DROP PARTITION SCHEME (partition_scheme_name);
Does this meet the goal?
A. Yes
B**. No**
Answer:B

Q- You have an Azure SQL database that contains a column named Notes.

A security review discovers that Notes contains sensitive data.

You need to ensure that the data is protected so that neither the stored values nor the query inputs reveal information about the actual data. The solution must prevent a user from inferring relationships or repetitions in the data based on
the encrypted output.
Which should you use?

A. **Always Encrypted with randomized encryption**

B. Always Encrypted with deterministic encryption

C. Always Encrypted with secure enclaves

D. Row-Level Security (RLS)

Answer: A

Q-You have an Azure SQL database that supports a customer-facing API. The API calls a stored procedure named dbo.GetCustomerOrders thousands of times per hour.
After a deployment that updated indexes and statistics, users report that the API endpoint backed by dbo.GetCustomer Orders is slower. In Query Store, the same query now has two persisted execution plans. During the last hour, the newer plan had a significantly higher average duration and CPU time than the older plan. You need to restore the previous performance quickly, without changing the API code.
Which Transact-SQL command should you run?
A. EXEC sys.sp_query_store_set_hints
B. DBCC FREEPROCCACHE
C. EXEC sp_query_store_force_plan
D. ALTER DATABASE
Answer: C
Q-Which tool helps orchestrate data pipelines for Al workflows?
A. Azure DevOps
B. **Azure Data Factory**
C. Power BI
D. Azure Monitor
Answer: B

Q- What is the primary purpose of Azure Al in SQL development?
A. Replace SQL Server
B. Automate infrastructure deployment
C. Enhance querying and analytics with Al capabilities
D. Eliminate the need for databases
Answer- C

Q-You need to recommend a solution for the TransactionProcessing application that meets the security requirements.
What should you include in the recommendation?
A. a service principal
B. a system-assigned managed identity
C. a shared access key
D. a user-assigned managed identity
Answer: B

Q- You have a GitHub Actions workflow that builds and deploys an Azure SQL database. The schema is

stored in a GitHub repository as an SDK-style SQL database project.
Following a code review, you discover that you need to generate a report that shows whether the production schema
has diverged from the model in source control.
Which action should you add to the pipeline?

A. SqlPackage.exe /Action: DriftReport

B. SqlPackage.exe /Action: Deploy Report

C. SqlPackage.exe /Action:Extract

D. SqlPackage.exe /Action:Script

Answer A

Q-Your development team uses Visual Studio Code with the MSSQL extension and GitHub Copilot Chat. The team connects to an Azure SQL database by using individual database logins and uses the @mssql chat participant to generate and run Transact-SQL queries from prompts.
What is used to ensure that GitHub Copilot Chat-generated queries run in the context of the developer?

A. **SQL permissions**

B. Azure RBAC permissions

C. GitHub organization permissions

D. Shared MCP instruction files

Answer: A

Q- You have an Azure container app named app1-contoso-001 that hosts a Data API Builder (DAB) container in front of an Azure SQL database. You add an entity named Todo that uses a source named dbo.todos. Which URL pattern should clients use to access the Todo REST endpoint?

A. https://app1-contoso-001.azurestaticapps.net/data-api/graphql/Todo

B. **https://app1-contoso-001.azurestaticapps.net/api/Todo**

C. https://app1-contoso-001.azurestaticapps.net/data-api/Todo

D. https://app1-contoso-001.azurestaticapps.net/data-api/api/Todo

Answer: B

Q- You implement Procedure Documents to support the planned changes. When users consume data through the Retrieval Augmented Generation (RAG) pattern, they experience data retrieval delays. You need to improve the data retrieval performance and reduce the number of tokens per retrieval. What should you implement?

A. **chunking**

B. embeddings

C. a small language model (SLM)

D. JSON content

Answer: A

You have a GitHub Enterprise subscription. Your team is developing an Azure SQL dataset solution from a locally cloned GitHub repository by using Visual Studio Code and GitHub Copilot Chat. A mix of GitHub Copilot instructions is configured at different levels, including organization-wide, repository-wide, agent-specific, and personal. Which instructions will take precedence over the others?

A. repository-wide

B. **personal**

C. organization-wide

D. agent-specific load PD
Answer: B

Q-What is a key benefit of embedding vectors in SQL databases?
A. Faster backups
B. Improved indexing
C. **Semantic search capability**
D. Reduced storage
Answer: C
Q- What is the role of Azure Synapse in AI + SQL solutions?
A. Data warehousing and analytics
B. Authentication
C. DNS management
D. Email services
Answer: A

Q-Which scenario best fits AI-enhanced SQL querying?
A. Static reports
B. Manual backups
C. **Chatbot querying database data**
D. File compression
Answer: C

Q-An e-commerce platform stores 25 million product embeddings in Azure SQL Database. Users report that semantic search queries are becoming slower as the dataset grows. The solution must improve search performance while minimizing application changes. Which action should be taken first?
A Replace semantic search with full-text search
B Increase embedding dimensions
C Optimize vector indexing and similarity search operations
D Move embeddings to CSV files in bob storage
Answer-C

Q- A healthcare provider stores patient records and vector embeddings used by an AI assistant. The organization must meet strict compliance requirements and ensure only authorized users can access sensitive information. Which combination of controls should be prioritized?
A Public access and caching
B Encryption, RBAC, and auditing
C Increased token limits
D Database replication only
Answer-B

Q- A company is building a customer-support assistant using Azure SQL Database and Azure OpenAI. Support articles are updated several times per day, and the organization requires AI-generated responses to reflect changes immediately without retraining the model. Which solution should you recommend?
A Fine-tune the Azure OpenAI model whenever articles change
B **Use RAG with embeddings stored in Azure SQL Database**
C Store all articles in application memory
D Increase the model temperature
Answer-B

Q- A financial services company has deployed an Al assistant powered by Azure OpenAl. Users occasionally receive responses containing information that does not exist in the organization's documentation. Management wants to reduce hallucinations while ensuring responses are based on approved data sources. What should the development team implement?

A Increase model temperature

B Fine-tune the model weekly

C Use RAG with trusted document retrieval

D Increase response length

Answer-C

Q- You need to generate embeddings to resolve the issues identified by the analysts. Which column should you use?

Options: A. vehicleLocation

B. i**ncidentDescrlption**

C. incidentType

D. SeverityScore

Answer: B

Q- You need to recommend a solution for the development team to retrieve the live metadata. The solution must meet the development requirements. What should you include in the recommendation? Options:

A. Export the database schema as a .dacpac file and load the schema into a GitHub Copilot context window.

B. Add the schema to a GitHub Copilot instruction file.

C. Use an MCP server

D. Include the database project in the code repository.

Answer: C

Q- You need to recommend a solution that will resolve the ingestion pipeline failure issues. Which two actions should you recommend? Each correct answer presents part of the solution. NOTE: Each correct selection is worth one point. Options:

A. Enable snapshot isolation on the database.

B. Use a trigger to automatically rewrite malformed JSON.

C. Add foreign key constraints on the table.

D. Create a unique index on a hash of the payload.

E. Add a check constraint that validates the JSON structure.

Answer: D, E

Q-You need to enable similarity search to provide the analysts with the ability to retrieve the most relevant health summary reports. The solution must minimize latency. What should you include in the solution? Options:

A. a computed column that manually compares vector values

B. a standard nonclustered index on the Fmbeddings (vector (1536)) column

C. a full-text index on the Fmbeddings (vector (1536)) column

D. a vector index on the Embedding* (vector (1536)) column

Answer: D

Q- You need to enable similarity search to retrieve the most relevant health summary reports while minimizing latency. The table already includes an 'Embeddings (vector(1536)) column. What should you include in the solution?
A. A standard nonclustered index on the vector column.
B. A full-text index on the vector column.
C. A vector index on the vector column.
D. A computed column that manually compares vector values.
Answer-C

Q- You have a RAG service that passes order details to a large language model (LLM). Which Transact-SQL command should you use to generate a nested array of line items within each order?
A. FOR JSON PATH
B. JSON_QUERY
C. JSON_VALUE
D. OPENJSON
Answer- B

Q - To minimize token usage when sending data to an LLM for grounding, which field should typically be omitted from the JSON context if it has no semantic value?
A. ProductName
B. FAQ Question
C. FAQ ID
D. FAQ Answer
Answer- C

Q- You need to call an Azure OpenAI REST endpoint from SQL Server 2025 with the highest level of security. The instance has managed identity enabled. What is the preferred authentication method?
A. Database scoped credential with an API key.
B. Database scoped credential with IDENTITY = 'Managed Identity'.
C. Shared Access Signature (SAS).
D. HTTP Header credentials with a rotated key.
Answer- B

Q- A database contains two million articles with embeddings that are updated frequently. Users report that semantic search results do not reflect updates until the next day. How should you ensure embeddings are updated while minimizing CPU usage on the database?
A. Create a trigger that calls AI_GENERATE_EMBEDDINGS for each row. B. Run an hourly job to regenerate all embeddings.
C. Enable Change Data Capture (CDC) and use an external Azure Function to process changes.
D. Modify the query to use VECTOR_DISTANCE instead of VECTOR_SEARCH.
Answer- C

Q- You are implementing a hybrid search that returns the top 20 results based on 60% vector distance and 40% full-text rank. Which function should you use to retrieve the exact distance for this formula?
A. VECTOR_SEARCH
B. VECTOR_DISTANCE
C. VECTOR_NORMALIZE
D. VECTORPROPERTY

Answer- B

Q- To filter a table with 120 million rows by a single calendar day efficiently (SARGable), which pattern should you use in your `WHERE' clause? A. WHERE CONVERT (char(10), CreateDate, 121) = @DateString
B. WHERE CreateDate >= @StartDate AND CreateDate < @EndDate
C. WHERE DATEDIFF (day, CreateDate, GETDATE()) = 0
D. WHERE YEAR(CreateDate) = 2024 AND MONTH (CreateDate)=1
Answer-B

Q- You need a reusable object that calculates a discount percentage and returns just one numeric result to the calling query. Which object fits best?
A. INSTEAD OF trigger
B. View
C. Scalar function
D. Stored procedure
Answe: C

Q- Which function should you use when you need a normalized version of a vector before comparison?
A. REGEXP_LIKE
B. VECTOR_DISTANCE
C. VECTORPROPERTY
D. VECTOR NORMALIZE
Answer- D

Q -API telemetry such as request rate, failures, and traces should be captured for the SQL-backed API service. Which Azure service is the best fit?
A. Application Insights
B. Change Tracking
C. Log Analytics
D. SEQUENCE
Answer- A

Q- An Azure-hosted application needs to connect to Azure SQL without storing usernames or passwords in configuration files. Which approach is best?
A. Service principal secret
B. Managed Identity
C. Password in connection string
D. Password in connection string
Answer- C

Q-You need to return a JSON array that contains several literal values in a fixed order. Which function should you use?
A. JSON_ARRAYAGG
B. JSON_ARRAY
C. JSON_OBJECT
D. OPENJSON
Answer- B

Q- You need current engine-state information such as wait statistics and cached query activity while

troubleshooting performance. Which source is best?
A. RRF
B. DMVS
C. Query Store
D. JSON_OBJECT
Answer- B

Q- You are deciding which chat session option to change when the current model cannot reliably return a machine-readable deployment manifest. What should you adjust first?
A. Structured output model
B. Partition switching
C. Application Insights
D. Row-Level Security
Answer-A

Q- You need tamper-evident history for records, while a separate requirement asks for who performed access and change operations. Which feature addresses the tamper- evident part?
A. Database auditing
B. Change Tracking
C. Dynamic Data Masking
D. Ledger table
Answer-D

Q- A client needs to request exactly the fields and relationships it wants from one endpoint rather than multiple fixed REST payloads. Which endpoint style is the best fit?
A. Change Tracking
B. GraphQL endpoint
C. Database auditing
D. REST endpoint
Answer-B

Q- A service account needs read access to one table and nothing else. Which approach is preferable?
A. db_datareader
B. Columnstore index
C. SQL authentication
D. Object-level permissions
Answer-D

Q- Contoso's ingestion pipeline sometimes fails due to malformed JSON and duplicate payloads. Which two actions should you recommend?
A. Enable snapshot isolation on the database.
B. Add a check constraint that validates the JSON structure.
C. Create a unique index on a hash of the payload.
D. Add foreign key constraints on the table.
Answer: B,C

Q- You have an Azure SQL database that contains a table named Rooms. Roomswas created by using the following Transact-SQL statement.

```
CREATE TABLE Rooms
(
    RoomID int PRIMARY KEY,
    Owner nvarchar(100),
    Capacity int
);
```

You discover that some records in the Rooms table contain NULL values for the Owner field.
You need to ensure that all future records have a value for the Owner field.
What should you add?
O A. a foreign key
O B. a check constraint
O C. a nonclustered index
O D. a unique constraint
Answer- B

Hotspot Question

Q- You have a SQL database in Microsoft Fabric that uses the default settings for a newly created database and contains a table named Sales.Orders. You have an application that uses two stored procedures to access Sales.Orders.
While monitoring database activity, you discover the following:
- sys.dm_exec_requests shows multiple sessions in a suspended state with wait_type = LCK_M_X. All the sessions show the same wait_resource, which maps to Sales.Orders, and the same nonzero blocking_session_id.
- sys.dm_exec_input_buffer(blocking_session_id, NULL) returns a last submitted command of BEGIN TRANSACTION UPDATE Sales.Orders.
- sys.dm_exec_sessions for the blocking session shows status = sleeping and open_transaction_count = 1.
For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
The blocking is caused by an uncommitted explicit transaction in the blocking session that is holding locks.	O	O
While an `UPDATE` operation on `Sales.Orders` is occurring, `SELECT` statements will be blocked.	O	O
Joining `sys.dm_tran_locks` to `sys.dm_exec_requests` will show which session holds locks involved in the blocking chain.	O	O

Q- Which service enables natural language querying over SQL data?
A. Azure Data Factory

B. Azure Blob Storage
C. Azure Synapse Analytics
D. Azure OpenAI Service
Answer- D

Q- Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have an SDK-style SQL database project stored in a Git repository. The project targets an Azure SQL database.
The CI build fails with unresolved reference errors when the project references system objects.
You need to update the SQL database project to ensure that dotnet build validates successfully by including the correct system objects in the database model for Azure SQL Database. Solution: Build the project by running dotnet build -bl -flp:v=diag.
Does this meet the goal?
A. No
B. Yes
Answer- A

Case Study 1

Existing Environment
Contoso has an Azure subscription in North Europe that contains the corporate infrastructure. The current infrastructure contains a Microsoft SQL Server 2017 database. The database contains the following tables.

Table names	Column names
CustomerFeedback	• FeedbackId (int) (primarykey) • FeedbackJson (nvarchar (max))
Fleets	• FleetId (int) (primarykey) • FleetName (nvarchar(100)) • Description (nvarchar(256))
MaintenanceEvents	• MaintenanceId (int) (primarykey) • VehicleId (int) • LastModifiedUTC (datetime2) • Description (nvarchar(256))
SupportTickets	• TicketId (int) (primarykey) • FleetId (int) • CreatedUtc (datetime2)
UserAccounts	• UserId (int) (primarykey) • UserPrincipalName (nvarchar(256)) • JobRole (nvarchar(256)) • StartDate (datetime2)
VehicleHealthSummary	• IncidentId (int) (primarykey) • VehicleId (int) • FleetId (int) • Summary (nvarchar(2000)) • LastUpdatedUtc (datetime2) • EngineStatus [bit] • EngineStatusLastUpdatedUtc (datetime2) • BatteryHealth (int) • Embeddings (vector (1536))

The FeedbackJson column has a full-text index and stores JSON documents in the following format.

```
{
  "text": "The battery drains too fast when driving uphill.",
  "category": "Battery",
  "metadata": {
    "appVersion": "5.2.1",
    "device": "Android",
    "language": "en-GB"
  }
}
```

The support staff at Contoso never has the UNMASKpermission.

Problem Statements
Contoso is deploying a new Azure SQL database that will become the authoritative data store for the following:

- AI workloads
- Vector search
- Modernized API access
- Retrieval Augmented Generation (RAG) pipelines

Sometimes the ingestion pipeline fails due to malformed JSON and duplicate payloads.
The engineers at Contoso report that the following dashboard query runs slowly.

```
SELECT VehicleId, LastUpdatedUtc, EngineStatus, BatteryHealth
FROM dbo.VehicleHealthSummary
WHERE FleetId = @FleetId
ORDER BY LastUpdatedUtc DESC;
```

You review the execution plan and discover that the plan shows a clustered index scan.
VehicleIncidentReportsoften contains details about the weather, traffic conditions, and location.
Analysts report that it is difficult to find similar incidents based on these details.
Requirements
Planned Changes
Contoso wants to modernize Fleet Intelligence Platform to support AI-powered semantic search over incident reports.
Security Requirements

Contoso identifies the following security requirements:
* Restrict the support staff from viewing Personally Identifiable Information (PII) data, which is full email addresses and phone numbers.
* Enforce row-level filtering so that analysts see only incidents for the fleets to which they are assigned.
The analysts can be assigned to multiple fleets.

Database Performance and Requirements
Contoso identifies the following telemetry requirements:
* Telemetry data must be stored in a partitioned table.
*Telemetry data must provide predictable performance for ingestion and retention operations.
* latitude, longitude, and accuracyJSON properties must be filtered by using an index seek.

Contoso identifies the following maintenance data requirements:
* Ensure that any changes to a row in the MaintenanceEventstable updates the corresponding value in the LastModifiedUtccolumn to the time of the change.
* Avoid recursive updates.
AI Search, Embeddings, and Vector Indexing
Contoso plans to implement semantic search over incident data to meet the following requirements:
* Embeddings must be stored in dedicated Azure SQL Database tables.
* Embeddings must be generated from rich natural language fields.
* Chunking must preserve semantic coherence.
* Hybrid search must combine the following:
- Vector similarity
-Keyword filtering or boosting

Development Requirements
The development team at Contoso will use Microsoft Visual Studio Code and GitHub Copilot and will retrieve live metadata from the databases. Contoso identifies the following requirements for querying data in the FeedbackJsoncolumn of the CustomerFeedbacktable:
Extract the customer feedback text from the JSON document.
* Filter rows where the JSON text contains a keyword.
* Calculate a fuzzy similarity score between the feedback text and a known issue description.
* Order the results by similarity score, with the highest score first.

Q- You need to recommend a solution that will resolve the ingestion pipeline failure issues.
Which two actions should you recommend? Each correct answer presents part of the solution. NOTE: Each correct selection is worth one point.

A. Add a check constraint that validates the JSON structure.
B. Add foreign key constraints on the table.
C. Create a unique index on a hash of the payload.
D. Enable snapshot isolation on the database.
E. Use a trigger to automatically rewrite malformed JSON.
Answer: A, C

Q -Which column should you use?
A. vehicleLocation
B. **incidentDescrlption** correct
C. incidentType
D. SeverityScore
Answer- B

Q- You have an Azure SQL database That contains a table named dbo.Products, dbo.Products contains three columns named Embedding Category, and Price. The Embedding column is defined as VECTOR(1536).
You use Ai_GENERME_EMBEDOINGS and VECTOR_SEARCH to support semantic search and apply additional filters on two columns named Category and Price.
You plan to change the embedding model from text-embedding-ada-002 to text-embedding-3-smalL
Existing rows already contain embeddings in the Embedding column.
You need to implement the model change. Applications must be able to use VECTOR_SEARCH without runtime errors.

What should you do first?

A. **Regenerate embeddings for the existing rows.**
B. Normalize the vector lengths before storing new embeddings.
C. Convert the Embedding column to nvacchar(mex).
D. Create a vector index on dbo.Products.Embedding.

Answer-A

Q- You need to recommend a solution that will resolve the ingestion pipeline failure issues.
Which two actions should you recommend? Each correct answer presents part of the solution. NOTE: Each correct selection is worth one point.

A. Enable snapshot isolation on the database.
B. Use a trigger to automatically rewrite malformed JSO

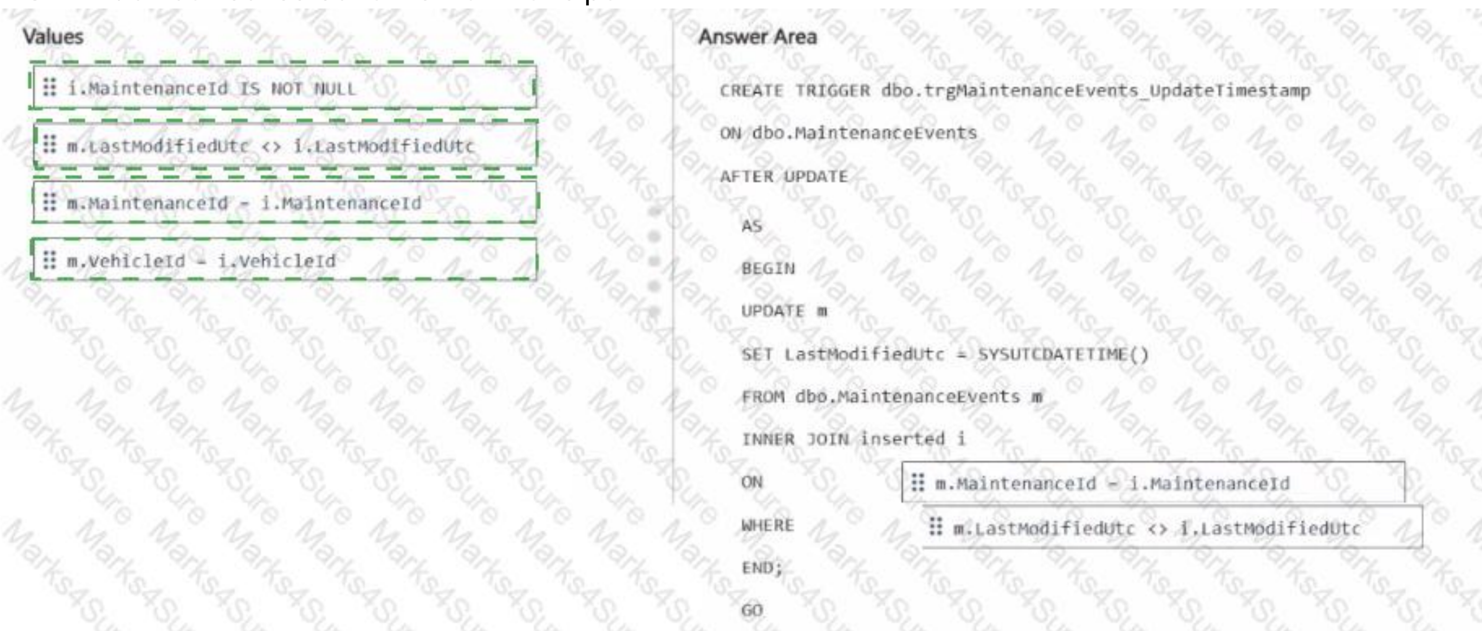
C. Add foreign key constraints on the table.correct

D. Create a unique index on a hash of the payload.correct

E. Add a check constraint that validates the JSON structure

Answer- D,E

Q- You need to meet the database performance requirements for maintenance data
How should you complete the Transact-SQL code? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



The correct drag-and-drop completion is:
ON m.maintenanceId = i.maintenanceId
WHERE m.LastModifiedUtc < > i.LastModifiedUtc

Q- You need to create a table in the database to store the telemetry data. You have the following Transact-SQL code.

```
CREATE TABLE dbo.VehicleTelemetry
(
  TelemetryId BIGINT IDENTITY(1,1) NOT NULL,
  VehicleId NVARCHAR(50) NOT NULL,
  TelemetryTimeUtc DATETIME2(3) NOT NULL,
  BatteryPercent TINYINT NULL,
  SpeedKmh SMALLINT NULL,
  LocationJson JSON NULL,
  ErrorCodesJson JSON NULL,
  RawPayload NVARCHAR(MAX) NULL,
  SysStartTime DATETIME2(7) GENERATED ALWAYS AS ROW START NOT NULL,
  SysEndTime DATETIME2(7) GENERATED ALWAYS AS ROW END NOT NULL,
  PERIOD FOR SYSTEM_TIME (SysStartTime, SysEndTime),
  CONSTRAINT PK_VehicleTelemetry PRIMARY KEY CLUSTERED (TelemetryId)
)
WITH
(
  SYSTEM_VERSIONING = ON
  (
    HISTORY_TABLE = dbo.VehicleTelemetryHistory
  )
);
GO
CREATE INDEX IX_VehicleTelemetry_Time ON dbo.VehicleTelemetry (TelemetryTimeUtc);
CREATE JSON INDEX JI_VehicleTelemetry_Location ON dbo.VehicleTelemetry (LocationJson) FOR
(
  '$.location. latitude', '$.location. longitude',
  '$.location. accuracy'
);
```

Answer Area

Statements	Yes	No
The code meets the database performance requirements for partitioning.	○	◉
The code meets the database performance requirements for JSON property querying.	◉	○
Queries that filter on `$.location.heading` will use the `JI_VehicleTelemetry_Location` index.	○	◉

Q- You need to recommend a solution to lesolve the slow dashboard query issue.
What should you recommend?

A. Create a clustered index on Lastupdatedutc.
B. On Fleetid, create a nonclustered index that includes Lastupdatedutc. inginestatus, and BatteryHealth.
C. On Lastupdatedutc. create a nonclustered index that includes Fleetid.
D. On Fleetid, create a filtered index where lastupdatedutc > DATEADD(DAV, -7, SYSuTCOATETIME()).

Q- You have an Azure SQL database named SalesDB that contains a table named dbo. Articles, dbo.Articles contains two million articles with embeddmgs. The articles are updated frequently throughout the day.
You query the embeddings by using VECTOR_SEARQi
Users report that semantic search results do NOT reflect the updates until the following day.

You need to ensure that the embeddings are updated whenever the articles change. The solution must minimize CPU usage on SalesDB

Which embedding maintenance method should you implement?

A Modify the query to use VECTOR.DTSTANCF instead of VECTOR.SEARCK

B. Enable change data capture (COC) on dbo.Articles and use an Azure Functions app to process CLX changes.

C Run an hourly Transact-SQL job that regenerates embeddings for all the rows in dbo.Articles.

D On dbo.Articles, create a trigger that calls AI GENERATE EMBEDOINGS for each inserted or updated row.

Answer- B

Q- You need to meet the development requirements for the FeedbackJson column

How should you complete the Transact SQL query? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answers:

- JSON_VALUE(f.FeedbackJson, ' $.text ') AS FeedbackText
- CONTAINS(FeedbackJson, @Keyword)
- SimilarityScore

Answer Area

```
SELECT
  f.FeedbackId,
  f.VehicleId,
  [dropdown] ,
```

- CONTAINS(FeedbackJson, @keyword)
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.details.comment'), @Keyword) < 3
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.text'), @Keyword) < 3
- JSON_QUERY(f.FeedbackJson, '$.text', @KnownIssueDescription) AS FeedbackText
- JSON_VALUE(f.FeedbackJson, '$.text') AS FeedbackText
- SimilarityScore

```
  dbo.CustomerFeedback f
  JSON_VALUE(f.FeedbackJson, '$.text'),
  @KnownIssueDescription
 ) AS SimilarityScore
FROM
 dbo.CustomerFeedback f
WHERE
 [dropdown]
```

- CONTAINS(FeedbackJson, @Keyword)
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.details.comment'), @Keyword) < 3
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.text'), @Keyword) < 3
- JSON_QUERY(f.FeedbackJson, '$.text', @KnownIssueDescription) AS FeedbackText
- JSON_VALUE(f.FeedbackJson, '$.text') AS FeedbackText
- SimilarityScore

[dropdown]

- CONTAINS(FeedbackJson, @Keyword)
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.details.comment'), @Keyword) < 3
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.text'), @Keyword) < 3
- JSON_QUERY(f.FeedbackJson, '$.text', @KnownIssueDescription) AS FeedbackText
- JSON_VALUE(f.FeedbackJson, '$.text') AS FeedbackText
- SimilarityScore

Hotspot Question

Q- You have an Azure SQL database that contains a table named stores, stores contains a column named description and a vector column named embedding.
You need to implement a hybrid search query that meets the following requirements:
• Uses full-text search on description for the keyword portion
• Returns the top 20 results based on a combined score that uses a weighted formula of 60% vector distance and 40% full-text rank
How should you configure the query components? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Answer Area

Semantic query operator/function:

- VECTOR_DISTANCE and order by distance ascending
- VECTOR_SEARCH with METRIC cosine and TOP_N
- VECTORPROPERTY to calculate the similarity between two vectors

Keyword retrieval operator/function:

- CONTAINSTABLE on description and return ranked matches
- FREETEXTTABLE on description for keyword scoring
- JSON_VALUE extraction from description for keyword scoring

Final ranking expression:

- order by (distance * 0.6) + ((1.0 RANK/1000.0) * 0.4)
- order by (distance * 0.6) + (RANK * 0.4)
- order by (distance + RANK), and then apply TOP 20

Answer Area

Semantic query operator/function: VECTOR_DISTANCE and order by distance ascending

Keyword retrieval operator/function: CONTAINSTABLE on description and return ranked matches

Final ranking expression: order by (distance * 0.6) + (RANK * 0.4)

Q- You have a SQL database in Microsoft Fabric named SalesDB that contains a table named dbo.Products. You need to modify SalesDB to meet the following requirements:

- Create a vector index on the appropriate column.
- Use a supplied natural language query vector.

How should you complete the Transact-SQL code? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Answer Area

```
CREATE VECTOR INDEX idx_products_embedding ON dbo.Products ( [dropdown] )
WITH (METRIC = 'cosine', TYPE = 'DiskANN');
DECLARE @query_vector VECTOR(1536) = @SuppliedVector
SELECT
    t.product_id,
    t.product_name,
    s.distance
FROM
    [dropdown] (
        TABLE = dbo.Products AS t,
        COLUMN = embedding,
        SIMILAR_TO = @query_vector,
        METRIC = 'cosine',
        TOP_N = 10
  ) AS s
[dropdown]
```

- distance
- embedding
- product_name

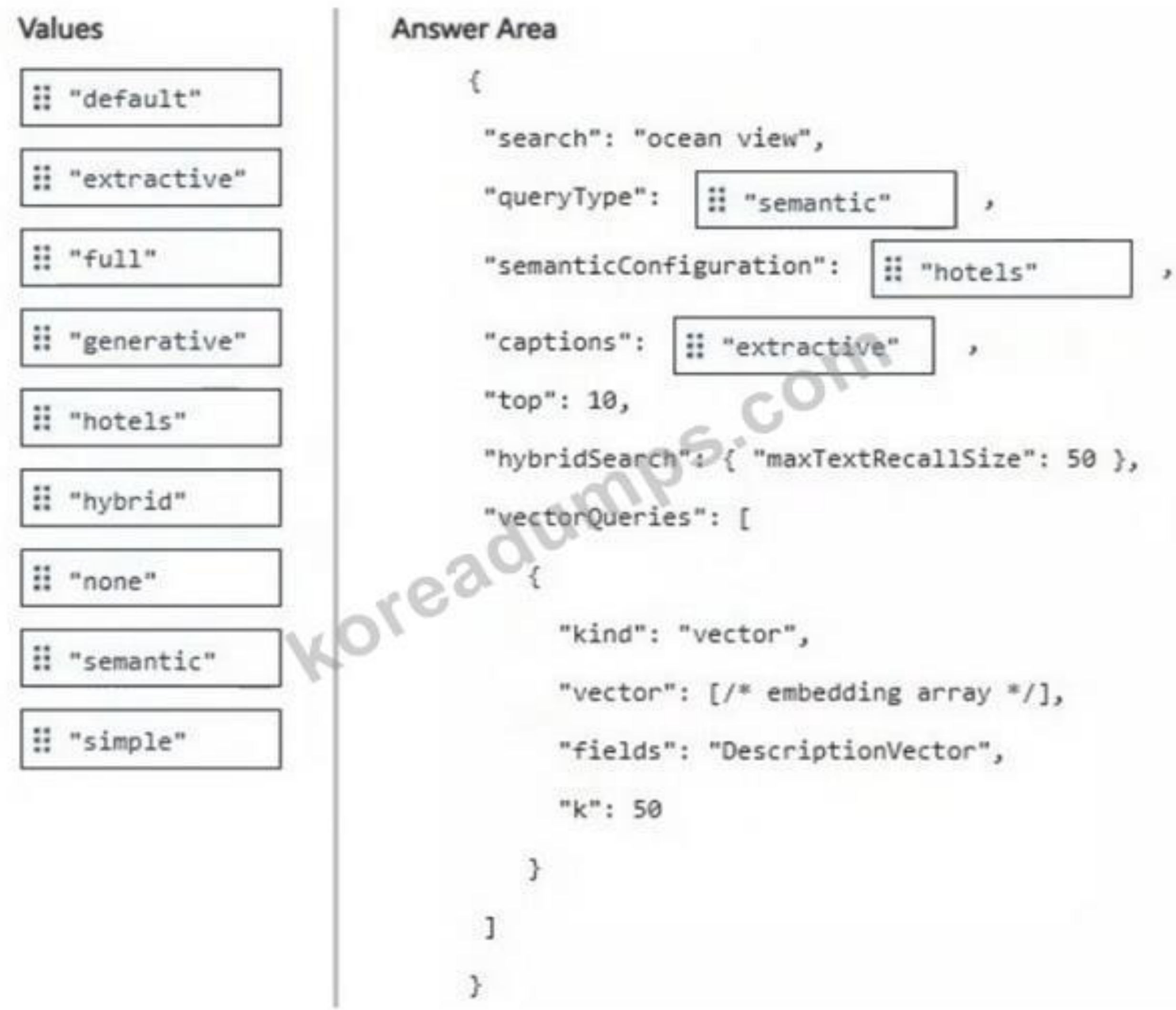
- VECTOR_DISTANCE
- VECTOR_NORMALIZE
- VECTOR_SEARCH

- GROUP BY s.distance
- ORDER BY s.distance
- PARTITION BY s.distance

Drag and Drop Question

Q- You have an Azure Al Search service and an index named hotels that includes a vector field named Description Vector. You query hotels by using the Search Documents REST API.
You need to implement a hybrid search query that uses DescriptionVector and includes captions.
How should you complete the REST request body? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar

between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



Q- Which component handles security in Azure SQL?
A. Azure CDN
B. Azure Functions
C. Azure Key Vault
D. Azure Al Studio
Answer- C

Q- Your development team uses GitHub Copilot Chat in Microsoft SQL Server Management Studio (SSMS) to generate and run Transact-SQL queries against an Azure SQL database named DB1. DB1 contains tables that store sensitive customer data.
You need to ensure that any Transact-SQL queries that run from GitHub Copilot Chat in SSMS are restricted by the same permissions as the developer's database login.
What prevents the GitHub Copilot Chat-run queries from accessing data beyond the developer's access?

A. GitHub Copilot Chat runs queries in a read-only sandbox that is isolated from production database permissions.
B. GitHub Copilot Chat uses different row-level security (RLS) policies than the developer.

C. GitHub Copilot Chat runs queries by using the developer's database identity and permissions.
D. GitHub Copilot Chat filters query results on the client side to remove rows the developer is unauthorized to see.

Hotspot Question
Q- You have an Azure SQL database that contains a table named Sales.Customer. Sales. Customer contains columns named Customerld, FullName, Email, TaxID, and Regionld. You have a database role named App Support that is used by a support application.
You need to implement a security solution for AppSupport that meets the following requirements:
-App Support must be prevented from viewing TaxID.
-AppSupport must be able to query Sales. Customer to troubleshoot issues.
- AppSupport must be able to run a stored procedure named
    Sales.usp_GetCustomerByCustomerld.

Which Transact-SQL statements should you include in the solution? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

To run Sales.usp_GetCustomerByCustomerId:

DENY SELECT ON OBJECT::Sales.Customer TO AppSupport;
DENY SELECT ON OBJECT::Sales.Customer(TaxID) TO AppSupport;
GRANT EXECUTE ON OBJECT::Sales.usp_GetCustomerByCustomerId TO AppSupport;
GRANT SELECT ON OBJECT::Sales.Customer TO AppSupport;
GRANT SELECT ON OBJECT::Sales.usp_GetCustomerByCustomerId TO AppSupport;
REVOKE SELECT ON OBJECT::Sales.Customer(TaxID) FROM AppSupport;

To query Sales.Customer:

DENY SELECT ON OBJECT::Sales.Customer TO AppSupport;
DENY SELECT ON OBJECT::Sales.Customer(TaxID) TO AppSupport;
GRANT EXECUTE ON OBJECT::Sales.usp_GetCustomerByCustomerId TO AppSupport;
GRANT SELECT ON OBJECT::Sales.Customer TO AppSupport;
GRANT SELECT ON OBJECT::Sales.usp_GetCustomerByCustomerId TO AppSupport;
REVOKE SELECT ON OBJECT::Sales.Customer(TaxID) FROM AppSupport;

To prevent viewing TaxID:

DENY SELECT ON OBJECT::Sales.Customer TO AppSupport;
DENY SELECT ON OBJECT::Sales.Customer(TaxID) TO AppSupport;
GRANT EXECUTE ON OBJECT::Sales.usp_GetCustomerByCustomerId TO AppSupport;
GRANT SELECT ON OBJECT::Sales.Customer TO AppSupport;
GRANT SELECT ON OBJECT::Sales.usp_GetCustomerByCustomerId TO AppSupport;
REVOKE SELECT ON OBJECT::Sales.Customer(TaxID) FROM AppSupport;

You havea GitHub Actions work flowt hat builds and deploys an Azure SQL database. The schema is stored in a GitHub repository as an SDK-style SQL database project. Following a code review, you discover that you need to generate a report that shows whether the production schema has diverged from the model in source control. Which action should you add to the pipeline?

A. SqlPackage.exe/Action:DriftReport
B. SqlPackage.exe/Action:Extract
C. SqlPackage.exe/Action:Script
D. SqlPackage.exe/Action:DeployReport

Answer: A
Explanation: Microsoft documents that Drift Report creates an XML report showing changes that have been made to the registered database since it was last registered . That is the action intended to detect

whether the production schema has diverged from the expected model base line in your deployment workflow. This is different from Deploy Report , which shows the changes that would be made by a publish action . In other words:
* DriftReport answers :Has the deployed database drifted from the registered state/model?
* DeployReport answers:What changes would be applied ifI published now?
The other options are not the right fit:
* Extract creates a DACPAC from an existing database, nota drift analysis report. * Script generates a deployment script, not a schema-drift report. So to generate a report that shows whether production has diverged from the model in source control,add: SqlPackage.exe/Action:DriftReport

Q- You have an Azure SQL database that contains tables named dbo.ProduetDocsand dbo. ProductuocsEnbeddings. dbo.ProductOocscontains product documentation and thefollowing columns:
* Docld (int)
* Title(nvdrchdr(200))
* Body (nvarthar(max))
* LastHodified (datetime2) The documentation is edited throughout the day.
dbo.ProductDocsEabeddingscontains the following columns:
* Dotid (int)
* ChunkOrder (int)
* ChunkText (nvarchar(aax))
* Embedding (vector(1536))
The current embedding pipeline runs once per night Vou need to ensure that embeddings are updated every time the underlying documentation content changes The solution must NOT ' equire a nightly batch process. What should you include in the solution?
A.a smaller embedding model
B.change tracking on dbo.ProductDocs
C. fixed-size chunking
D. table triggers

Answer: B

Q- You have a SQL database in Microsoft Fabric that contains the following functions:
* A multi-statement table-valued function (TVF) named sales.mstvf_orderStatus() that returns order status information
* A scalar user-defined function (UOF) named dbo.ufn_GetlaxMultiplier(

Drag and Drop Question
Q- You have an Azure SQL database that contains a table named dbo.Customers. dbo.Customers contains the following columns:
- CustomerId (int)(primary key)
- ProfileJson (nvarchar(max))
You have an application that returns phone numbers in a format of +000 000-000-0000. The phone numbers are stored in ProfileJson.
You need to write a query that returns:
- One row per customer
- A PhoneNUmerals column that contains only the digits
How should you complete the Transact-SQL query? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar

between panes or scroll to view content.
NOTE: Each correct selection is worth one point.

Values

- JSON_QUERY(c.ProfileJson, '$.contact.phone')
- JSON_VALUE(c.ProfileJson, '$.contact.phone')
- OPENJSON(c.ProfileJson, '$.contact.phone')
- p.PhoneRaw
- REGEXP_LIKE(p.PhoneRaw, '[^0-9]', '')
- REGEXP_REPLACE(p.PhoneRaw, '[^0-9]', '')
- REGEXP_SUBSTR(p.PhoneRaw, '[0-9]+')

Answer Area

```
WITH PhoneCTE AS
(
    SELECT DISTINCT
        c.CustomerId,
        [JSON_VALUE(c.ProfileJson, '$.contact.phone')]  AS PhoneRaw
    FROM dbo.Customers AS c
)
SELECT
    p.CustomerId,
    [REGEXP_REPLACE(p.PhoneRaw, '[^0-9]', '')]  AS PhoneNUmerals
FROM PhoneCTE AS p
WHERE     [p.PhoneRaw]     IS NOT NULL;
```


Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

Q- You have an Azure SQL database named SalesDB. SalesDB contains a table named dbo.Articles. dbo.Articles contains the following columns:
- ArticleId
- Title
- Body
- LastModifiedUtc
- EmbeddingVector

You have an application that generates embeddings from the concatenation of Title and Body and stores the results in EmbeddingVector.
You plan to implement an incremental embedding maintenance method that will use change data capture (CDC) to update embeddings only for rows that change, without scanning the entire table.
You need to ensure that only the columns required to generate the embeddings are captured.
The solution must support querying net changes.
How should you complete the Transact-SQL script? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.

Values

- 0
- 1
- EXEC sys.sp_cdc_disable_db;
- EXEC sys.sp_cdc_enable_db;
- N'ArticleId, Title, Body'
- N'Title, Body, LastModifiedUtc, EmbeddingVector'

Answer Area

```
USE [SalesDB];
GO
[EXEC sys.sp_cdc_enable_db;]
GO
EXEC sys.sp_cdc_enable_table
    @source_schema = N'dbo',
    @source_name = N'Articles',
    @role_name = NULL,
    @captured_column_list = [N'ArticleId, Title, Body'] ,
    @supports_net_changes = [1] ;
GO
```


Hotspot Question

Q- You have an Azure SQL database that contains a table named Table1. Table1 contains 25,000,000 rows of data and a datetime2 column named DateKey. The data in Table1 spans the years 2020 through 2021.

You need to partition the data in Table1 by year. The solution must minimize how long it takes to rebuild or reindex the table.

How should you complete the Transact-SQL code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
CREATE PARTITION FUNCTION PartitionByYear (datetime2)
AS [ ▼ ]
FOR VALUES ( [ ▼ ] );
```

AS options:

- PARTITION
- RANGE LEFT
- RANGE RIGHT

FOR VALUES options:

- '2019-01-01 00:00:00', '2020-01-01 00:00:00', '2021-12-31 23:59:59'
- '2019-12-31 00:00:00', '2020-12-31 23:59:59'
- '2020-01-01 00:00:00', '2020-02-01 00:00:00', '2020-03-01 00:00:00'
- '2020-01-01 00:00:00', '2021-01-01 00:00:00'

Q- You have a Microsoft SQL Server 2025 database that contains a table named dbo.CustomerMessages. dbo.CustomerMessages contains two columns named MessageID (int) and MessageRaw (nvarchar(max)).

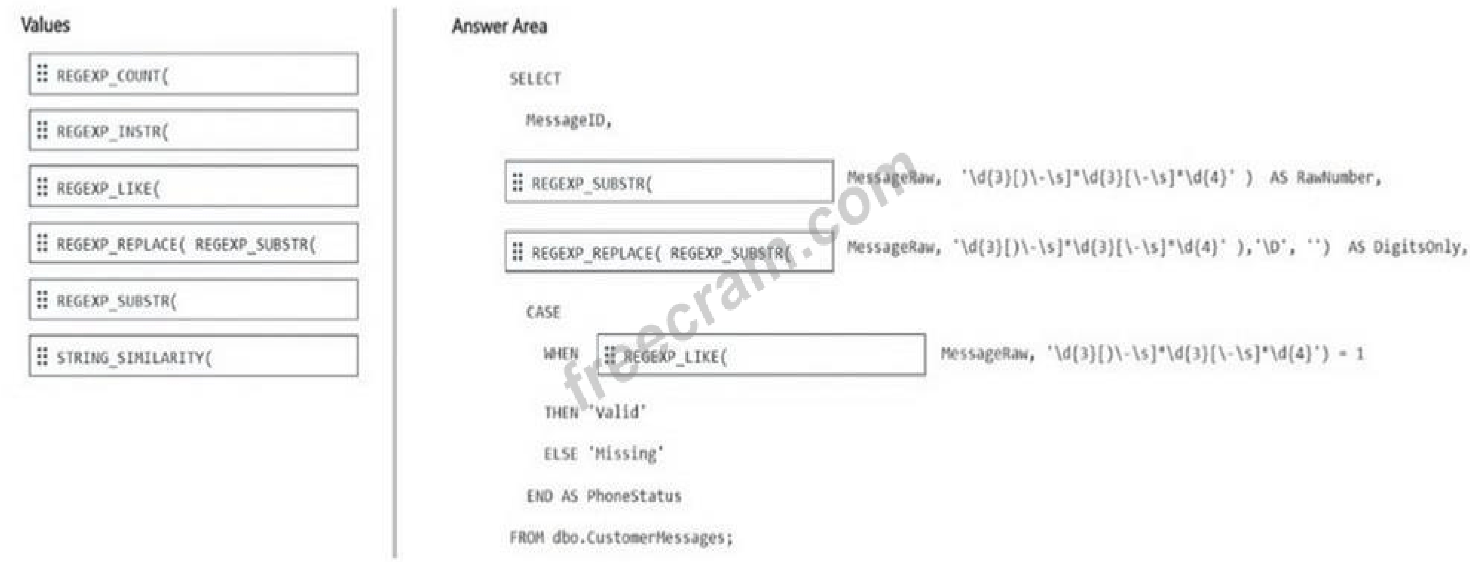
MessageRaw can contain a phone number in multiple formats, and some rows do NOT contain a phone number.

You need to write a single SELECT query that meets the following requirements:

- The query must return MessageID, RawNumber, DigitsOnly, and PhoneStatus.
- RawNumber must contain the first substring that matches a phone-number pattern, or NULL if no match exists.
- DigitsOnly must remove all non-digit characters from RawNumber, or return NULL.
- PhoneStatus must return valid when a phone number exists in MessageRaw, otherwise return Missing.

How should you complete the Transact-SQL query? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.



Q- You have a SQL database in Microsoft Fabric that contains a column named Payload. Payload stores customer data in JSON documents that have the following format.

```
{
"date": "2020-01-25",
"customer_email": "user@contoso.com",
...
}
```

Q- Data analysis shows that some customers have subaddressing in their email address, for example, user1+promo@contoso.com.

You need to return a normalized email value that removes the subaddressing, for example, user1 +promo@contoso.com must be normalized to user1@contoso.com.

Which Transact-SQL expression should you use?

A. REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+.*', '')

B. REGEXP_SUBSTR(JSON_VALUE(Payload, '$.customer_email'), '^[^+]+@.*$=')

C. REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+.*@', '@')

D. REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+.*$', '')

Answer C

Q-What is the benefit of using embeddings over keyword search?

A. Exact matches only
B. Smaller storage
C. Context-aware search
D. D. Faster deletes
Answer- C

Q- After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.
You have a SQL database in Microsoft Fabric that contains a table named dbo.Orders.
dbo.Orders has a clustered index, contains three years of data, and is partitioned by a column named OrderDate by month.
You need to remove all the rows for the oldest month. The solution must minimize the impact on other queries that access the data in dbo.Orders.
Solution: Run the following Transact-SQL statement.
DELETE FROM dbo.Orders
WHERE OrderDate < DATEADD(month, -36, SYSUTCDATETIME());
Does this meet the goal?
A. No **B.** Yes
Answer- A

Q- You have an Azure SQL database named AdventureWorksDB that contains a table named dbo.Employee.
You have a C# Azure Functions app that uses an HTTP-triggered function with an Azure SQL input binding to query dbo.Employee.
You are adding a second function that will react to row changes in dbo.Employee and write structured logs.
You need to configure AdventureWorksDB and the app to meet the following requirements:
* Changes to dbo.Employee must trigger the new function within five seconds.
* Each invocation must processes no more than 100 changes.
Which two database configurations should you perform? Each correct answer presents part of the solution. NOTE: Each correct selection is worth one point.
A Create an AFTER trigger on dbo.Employee for Data Manipulation Language (DML).
B SetSql Trigger MaxBatchSize to 100.
C Enable change tracking on the dbo. Employee table.
D Enable change tracking at the database level.
E Set Sql_Trigger_PollingIntervalMs to 5000.
F Enable change data capture (CDC) for dbo.Employee table changes

Answer: C, E

Q- You have a SQL database in Microsoft Fabric that contains a table named dbo.Orders, dbo.Orders has a clustered index, contains three years of data, and is partitioned by a column named OrderDate by month.
You need to remove all the rows for the oldest month. The solution must minimize the impact on other queries that access the data in dbo.orders.
Solution: Identify the partition number for the oldest month, and then run the following Transact-SQL statement.
statement.
TRUNCATE TABIE dbo.Orders

WITH (PARTITIONS (partition number));
Does this meet the goal?
A Yes B No
Answer- A

Q- You have an SDK-style SQL database project stored in a Git repository. The project targets an Azure SQL database.

The CI build fails with unresolved reference errors when the project references system objects.
You need to update the SQL database project to ensure that dotnet build validates successfully by including the correct system objects in the database model for Azure SQL Database.
Solution: Add the Microsoft.SqlServer.Dacpacs.Azure.Master NuGet package to the project.
Does this meet the goal?
A yes BNo
Answer-A

Q- You have an Azure SQL database that contains a column named Notes.
A security review discovers that Notes contains sensitive data.
You need to ensure that the data is protected so that neither the stored values nor the query inputs reveal information about the actual data. The solution must prevent a user from inferring relationships or repetitions in the data based on the encrypted output
Which should you use?
A Always Encrypted with secure enclaves
B Always Encrypted with randomized encryption
C row-level security <RLS)
D Always Encrypted with deterministic encryption
Answer -B

Q- You have an Azure SQL database That contains a table named dbo.Products, dbo.Products contains three columns named Embedding Category, and Price. The Embedding column is defined as VECTOR(1536).
You use Ai_GENERME_EMBEDOINGS and VECTOR_SEARCH to support semantic search and apply additional filters on two columns named Category and Price.
You plan to change the embedding model from text-embedding-ada-002 to text-embedding-3-smalL Existing rows already contain embeddings in the Embedding column.
You need to implement the model change. Applications must be able to use VECTOR_SEARCH without runtime errors.
What should you do first?
A. Regenerate embeddings for the existing rows.
B. Normalize the vector lengths before storing new embeddings.
C. Convert the Embedding column to nvacchar(mex).
D. Create a vector index on dbo.Products.Embedding.
Answer- A

Q- Your development team uses GitHub Copilot Chat in Microsoft SQL Server Management Studio (SSMS) to generate and run Transact-SQL queries against an Azure SQL database named DB1 DB1 contains tables that store sensitive customer data.
You need to ensure that any Transact SQL queries that run from GitHub Copilot Chat In SSMS are restricted by the same permissions as the developer's database login.

What prevents the GitHub Copilot Chat-run queries from accessing data beyond the developer's access?
A GitHub Copilot Chat runs queries in a read-only sandbox that is isolated from production database permissions.
B **GitHub Copilot Chat runs queries by using the developer's database identity and permissions.**
C GitHub Copilot Chat filters query results on (he client side to remove rows the developer is unauthorized to see.
D GitHub Copilot Chat uses different row-level security (RLS) policies than the developer.
Answer- B

Q- You have a SQL database in Microsoft Fabric that contains the following functions:
A multi-statement table-valued function (TVF) named sales.mstvf_orderStatus() that returns order status information
* A scalar user-defined function (UOF) named dbo.ufn_GetlaxMultiplier(^rax/Wt money, gStateCode char (2)) that returns a numeric multiplier used in tax calculations Reporting queries frequently join Sales.istvf_OrderStatus() to a table named Sales.SalesOrderHeader and return large result sets. A performance review shows that the queries produce inconsistent execution plans.
During a code review, a developer discovers that the following Transact-SQL statement produced an error.
EXEC @ret ufn_GetTaxMultiplier @TaxAmt 100.00, @StateCode ='WA';

```
EXEC @ret = ufn_GetTaxMultiplier @TaxAmt = 100.00, @StateCode = 'WA';
```

Answer Area

Statements	Yes	No
You can use GETDATE() in dbo.ufn_GetTaxMultiplier to produce nondeterministic results.	○	○
Rewriting Sales.mstvf_OrderStatus() as an inline table TVF will reduce the number of inconsistent execution plans.	○	○
Replacing ufn_GetTaxMultiplier with dbo.ufn_GetTaxMultiplier in the EXEC function statement will resolve the error.	○	○

The first statement is No. GETDATE() is a nondeterministic built-in function, and SQL Server UDF guidance does not support using nondeterministic built-ins inside a T-SQL user-defined function for this purpose. Microsoft explicitly classifies GETDATE() as nondeterministic.
The second statement is Yes. Multi-statement TVFs are a common source of weak cardinality estimates and unstable plans because the optimizer has less information about their result sizes. Microsoft documents features such as interleaved execution for multi-statement TVFs, which exist specifically because MSTVFs have optimization challenges, and Microsoft also notes that scalar/UDF style constructs can lead to poorer optimization behavior than relationally inlined alternatives. Rewriting an MSTVF as an inline TVF generally gives the optimizer better visibility and more consistent plans.
The third statement is Yes. Scalar functions must be invoked using at least the two-part name < schema > . < function > . Microsoft's documentation is explicit on that point for executing user-defined functions. So changing ufn_GetTaxMultiplier to dbo.ufn_GetTaxMultiplier resolves the naming error in the EXEC statement.

Q- You have a database named db1. The schema is stored in a Git repository as an SDK-style SQL database project.

The repository Contains the following GitHub Action workflow.

```
name: Database CI/CD
on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main
jobs:
  build-and-deploy:
    runs-on: ubuntu-latest
    steps:
    - name: Checkout code
      uses: actions/checkout@v3
    - name: Setup .NET
      uses: actions/setup-dotnet@v3
      with:
        dotnet-version: '8.x'
    - name: Build
      run: dotnet build db1.sqlproj --configuration Release
    - name: Deploy
      run: |
        SqlPackage /Action:Publish \
          /SourceFile:./bin/Release/db1.dacpac \
          /TargetConnectionString:"${{ secrets.Target_Connection_String }}"
  unit-tests:
    runs-on: ubuntu-latest
    needs: build-and-deploy
    if: github.ref == 'refs/heads/main'
    steps:
    - name: Checkout code
      uses: actions/checkout@v3
    - name: Setup .NET
      uses: actions/setup-dotnet@v3
      with:
        dotnet-version: '8.x'
    - name: Run unit tests
      run: dotnet test UnitTests.csproj --configuration Release
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No. NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
Unit tests run automatically whenever changes are pushed to `main`.	○	○
Schema validation occurs during the `Build` step.	○	○
Schema validation occurs during the `Deploy` step.	○	○

Answer-

Unit tests run automatically whenever changes are pushed to main. → **Yes**
Schema validation occurs during the Build step. → **Yes**
Schema validation occurs during the Deploy step. → **No**

Q- You need to recommend a solution for the development team to retrieve the live metadata. The solution must meet the development requirements.
What should you include in the recommendation?

A. Use an MCP server

B. Export the database schema as a .dacpac file and load the schema into a GitHub Copilot context window.

C. Add the schema to a GitHub Copilot instruction file.

D. Include the database project in the code repository.

Answer:A

Q- You have an Azure SQL database That contains database-level Data Definition Language (DDL) triggers, including a trigger named ddl_Audit.
You need to prevent ddl_Audit from firing during the next deployment. The trigger object must remain in place. Which Transact-SQL statement should you use?

A. DISABLE TRIGGER

B. ALTER TRIGGER

C. ALTER DATABASE

D. ALTER DATABASE AUDIT SPECIFICATION

E. ALTER SERVER AUDIT SPECIFICATION

Answer- A

Q- You have an Azure SQL database that contains the following SQL graph tables:
* A NODE table named dbo.Person
* An EDGE table named dbo.Knows
Each row in dbo.Person contains the following columns:
* Personid (int)
* DisplayName (nvarchar(100))
You need to use a HATCH operator and exactly two directed Knows relationships to return the Personid and DisplayName of people that are reachable from the person identified by an input parameter named @startPersonid.
Which Transact-SQL query should you use?

A)

```
SELECT p2.PersonId, @StartPersonId
FROM dbo.Person AS p1, dbo.Knows AS k1, dbo.Person AS p2, dbo.Knows AS k2, dbo.Person AS p3
WHERE p1.DisplayName = p2.DisplayName
  AND MATCH(p1-(k1)->p2-(k2)->p3);
```

```
SELECT p3.PersonId, p3.DisplayName FROM dbo.Person AS p1
JOIN dbo.Knows AS k1 ON 1 = 1
JOIN dbo.Person AS p2 ON 1 = 1
JOIN dbo.Knows AS k2 ON 1 = 1
JOIN dbo.Person AS p3 ON 1 = 1
WHERE p1.PersonId = @StartPersonId
  AND MATCH(p3<-(k2)-p2<-(k1)-p1);
```

B)

```
SELECT p3.PersonId, p3.DisplayName
FROM dbo.Person AS p1, dbo.Knows AS k1, dbo.Person AS p2, dbo.Knows AS k2, dbo.Person AS p3
WHERE p1.PersonId = @StartPersonId
  AND MATCH(p1-(k1)->p2)   AND MATCH(p2-(k2)->p3);
```

C)

```
SELECT p3.PersonId, p3.DisplayName
FROM dbo.Person AS p1, dbo.Knows AS k1, dbo.Person AS p2, dbo.Knows AS k2, dbo.Person AS p3
WHERE p1.PersonId = @StartPersonId
  AND MATCH(p1-(k1)->p2-(k2)->p3);
```

D)

A Option A
B Option B
C Option C
D **Option D**
Answer- D

Q- You have an Azure SQL database That contains a table named dbo.Products, dbo.Products contains three columns named Embedding Category, and Price. The Embedding column is defined as VECTOR(1536).
You use Ai_GENERME_EMBEDOINGS and VECTOR_SEARCH to support semantic search and apply additional filters on two columns named Category and Price.
You plan to change the embedding model from text-embedding-ada-002 to text-embedding-3-smalL Existing rows already contain embeddings in the Embedding column.
You need to implement the model change. Applications must be able to use VECTOR_SEARCH without runtime errors.
What should you do first?
A Regenerate embeddings for the existing rows.
B Normalize the vector lengths before storing new embeddings.
C Convert the Embedding column to nvacchar(mex).
D Create a vector index on dbo.Products.Embedding.
Answer- A

Q- You have an Azure SQL database.
You deploy Data API builder (DAB) to Azure Container Apps by using the mcr.nicrosoft.com/azure-databases/data-api-builder:latest image.
You have the following Container Apps secrets:
* MSSQL_COMNECTiON_STRrNG that maps to the SQL connection string
* DAB_C0HFT6_BASE64 that maps to the DAB configuration
You need to initialize the DAB configuration to read the SQL connection string.
Which command should you run?

A. dab init --database-type mssql --connection-string 'secretref:DAB_CONFIG_BASE64' --host-mode Production --config dab-config.json

B. dab init --database-type mssql --connection-string '@env('MSSQL_CONNECTION_STRING')' --host-mode Production --config dab-config.json

C . dab init --database-type mssql --connection-string 'secretref:mssql-connection-string' --host-mode Production --config dab-config.json

D .dab init --database-type mssql --connection-string '@env('DAB_CONFIG_BASE64')' --host-mode Production --config dab-config.json

Answer- B

Q- You have a Microsoft SQL Server 2025 instance that contains a database named SalesDB SalesDB supports a Retrieval Augmented Generation (RAG) pattern for internal support tickets. The SQL Server instance runs without any outbound network connectivity.
You plan to generate embeddings inside the SQL Server instance and store them in a table for vector similarity queries.
You need to ensure that only a database user account named AlApplicationUser can run embedding generation by using the model.
Which two actions should you perform? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A-Grant the control permission on SalesDB to AlApplicationUser.

B-Create a database audit specification on SalesDB owned by AlApplicationUser.

C-**Grant the execute permission on the external model project to AlApplicationUser.**

D-Create an external model project by using ONNX runtime and local paths.

E- Create an external model project that points to a Microsoft Foundry REST endpoint.

Answer- C,D

Q- You have an Azure SQL database that contains a table named dbo.ManualChunks.
dbo.HonualChunks contains product manuals
A retrieval query already returns the top five matching chunks as nvarchar(max) text.
You need to call an Azure OpenAI REST endpomt for chat completions. The request body must include both the user question and theretiieved chunks.
You write the following Transact-SQL code.

```
01 CREATE DATABASE SCOPED CREDENTIAL AzureOpenAIHeaders
02 WITH IDENTITY = 'HTTPEndpointHeaders',
03   SECRET = N'{"api-key":"<YOUR_AZURE_OPENAI_API_KEY>"}';
04 GO
05 CREATE OR ALTER PROCEDURE dbo.AskManuals
06   . . .
07   SELECT @chunks =
08   (
09     SELECT TOP (5)
10       mc.ChunkText AS [text]
11     FROM dbo.ManualChunks AS mc
12     ORDER BY mc.Score DESC
13     FOR JSON PATH
14   );
15   SET @payload =
16   (
17     SELECT
18       'system' AS [messages[0].role],
19       'Use only the provided manual chunks.' AS [messages[0].content],
20       'user' AS [messages[1].role],
21       CONCAT(@question, CHAR(10), JSON_QUERY(@chunks)) AS [messages[1].content]
22
23   );
24
25   EXEC @retval =
26     . . .
27 END;
28 GO
```

What should you insert at line 22?

A. FOR XHL AUTO, TYPE, XML SCHEMA,
B. FOR 3SON AUTO, IMCLUDE_MULL_VAIUES
C. FOR XHL PATH, INCLUDE_NULL_VAIUES
D. FOR JSON PATH, WITHOUT_ARRAY_WRAPPER

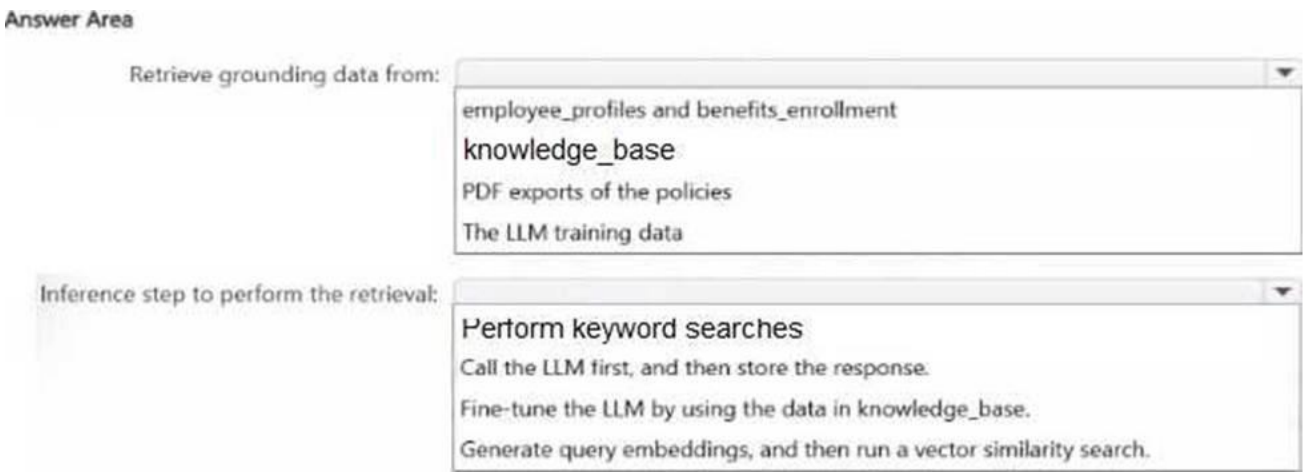
Answer- D : FOR JSON PATH, WITHOUT_ARRAY_WRAPPER.

Q- You have an Azure SQL database that contains a table named knowledgebase, knowledgebase stores human resources (HR) policy documents and contains columns named title, content, category, and embedding.

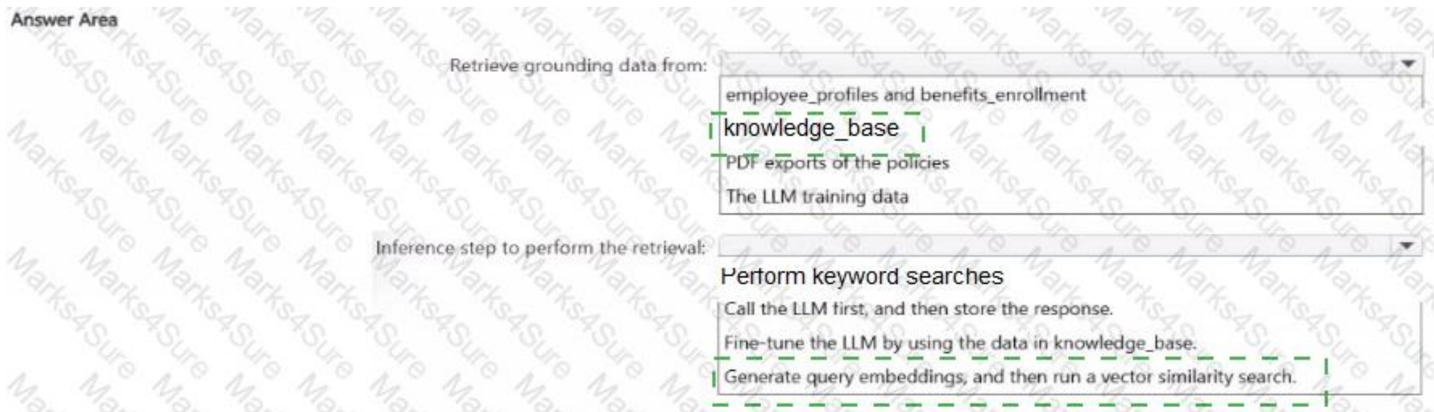
You have an application named App1. App1 queries two relational tables named employee_pnofiles and benefits_enrollnent that contain HR data. App1 hosts a chatbot that calls a large language model (LLM) directly.

Users report that the chatbot answers general HR questions correctly but provides outdated or incorrect answers when policies change. The chatbot also fails to answer questions that reference internal policy documents by title or category.

You need to recommend a Retrieval Augmented Generation (RAG) solution to resolve the chatbot issues. What should you recommend? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.



Or



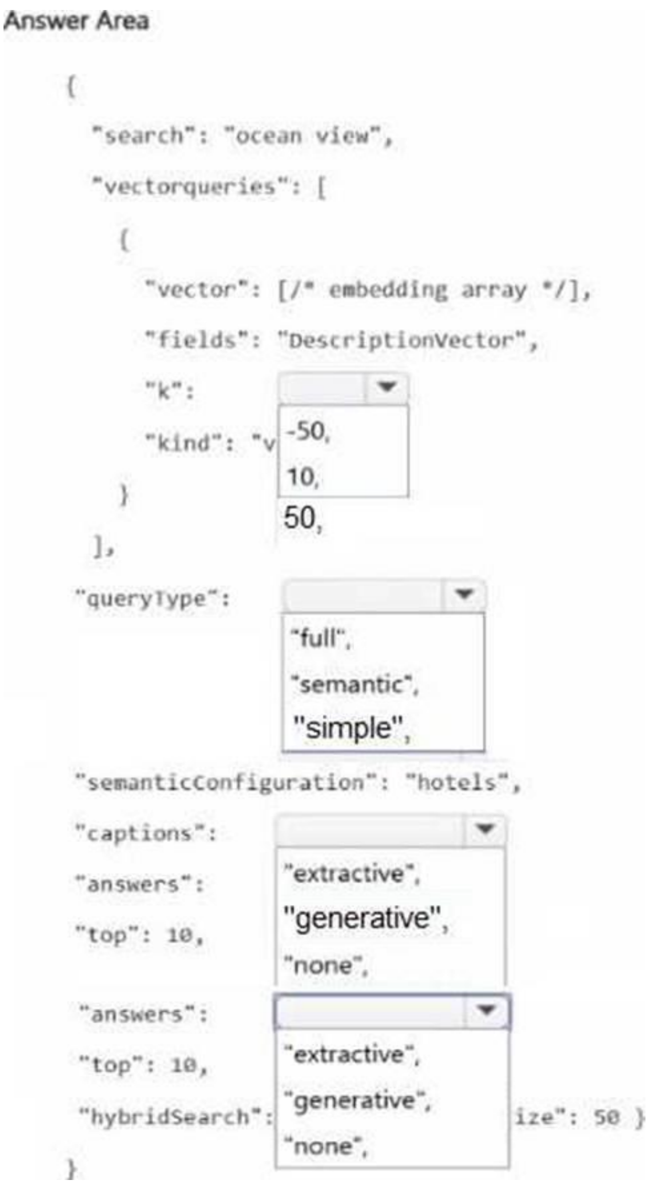
- 

Answer- correct recommendation is to retrieve grounding data from **knowledge_base** and, at inference time, **generate query embeddings and run a vector similarity search**.

Q- You need to design a generative Al solution that uses a Microsoft SOL Server 2025 database named DB1 as a data source.The solution must generate responses that meet the following requirements:
• Ait' grounded In the latest transactional and reference data stored in D61
• Do NOT require retraining or fine-tuning the language model when the data changes
• Can include citations or references to the source data used in the response
Which scenario is the best use case for implementing a Retrieval Augmented Generation (RAG) pattern? More than one answer choice may achieve the goal. Select the BEST answer
A. summarizing free-form user input text
B. training a custom language model on historical database data
C. answering user questions based on company-specific knowledge
D. generating marketing slogans based on user sentiment analysis
Answer- C

Q- You have an Azure Al Search service and an index named hotels that includes a vector Held named DescriptionVector.You query hotels by using the Search Documents REST API.

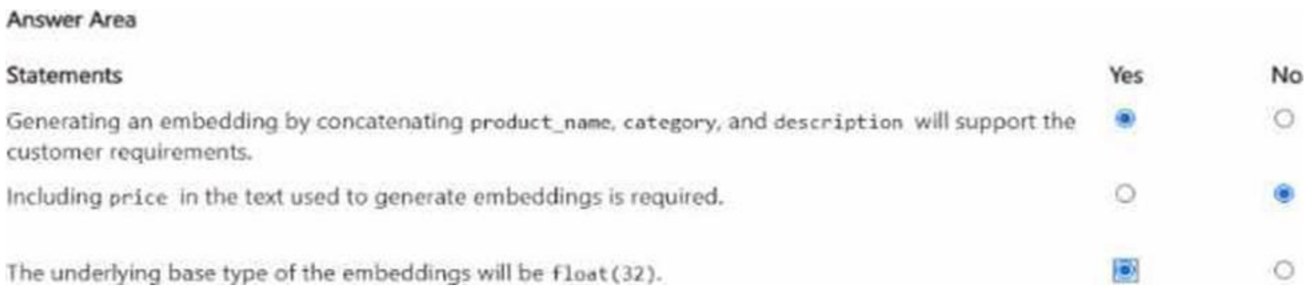
You add semantic ranking to the hybrid search query and discover that some queries return fewer results than expected, and captions and answers are missing.
You need to complete the hybrid search request to meet the following requirements:
• Include more documents when ranking.
• Always include captions and answers.



k = 50
queryType = "semantic"
captions = "extractive"
answers = "extractive"

Q- Your company has an ecommerce catalog in a Microsoft SQL Server 202b database named SalesDB SalesDB contains a table named products, products contains the following columns:
• product.id (int)
• product_name (nvarchar(200))
• description (nvarchar(max))
• category (nvarchar(50))
• brand (nvarchar(W))
• price (decimal)
• sku (nvarchar(40))
The description fields ate updated dairy by a content pipeline, and price can change multiple times per day. You want customers to be able to submit natural language queries and apply structured filters for

brand and price. You plan to store embeddings in a new VECTOR(1536) column and use VECTOR_SEARCH(... METRIC=' cosine' ...).
For each of the following statements, select Yes if the statement is true. Otherwise, select No. NOTE: Each correct selection is worth one point.

- 

Answer Area

Statements	Yes	No
Generating an embedding by concatenating `product_name`, `category`, and `description` will support the customer requirements.	●	○
Including `price` in the text used to generate embeddings is required.	○	●
The underlying base type of the embeddings will be `float(32)`.	●	○

Q- You have a Microsoft SQL Server 2025 instance that contains a database named SalesDB SalesDB supports a Retrieval Augmented Generation (RAG) pattern for internal support tickets. The SQL Server instance runs without any outbound network connectivity.
You plan to generate embeddings inside the SQL Server instance and store them in a table for vector similarity queries.
You need to ensure that only a database user account named AlApplicationUser can run embedding generation by using the model.
Which two actions should you perform? Each correct answer presents part of the solution. NOTE: Each correct selection is worth one point.

A. **Grant the control permission on SalesDB to AlApplicationUser.**
B. Create a database audit specification on SalesDB owned by AlApplicationUser.
C. **Grant the execute permission on the external model project to AlApplicationUser.**
D. **Create an external model project by using ONNX runtime and local paths.**
E. Create an external model project that points to a Microsoft Foundry REST endpoint.

Answer-A,C,D (option A is not correct probably)

Q-You have an Azure SQL database that contains the following tables and columns.

Tables	Columns
`Articles`	• `ArticleId (int)` • `Title (nvarchar(200))` • `Notes (nvarchar(max))` • `Description (nvarchar(max))`
`NotesEmbeddings`	• `ArticleId (int)` • `ChunkIndex (int)` • `Embedding (vector(1536))`
`DescriptionEmbeddings`	• `ArticleId (int)` • `ChunkIndex (int)` • `Embedding (vector(1536))`

Embeddings in the NotesEnbeddings and DescriptionEabeddings tables have been generated from values in the Description and notes columns of the Articles table by using different chunk sizes.
You need to perform approximate nearest neighbor (ANN) queries across both embedding tables.
The solution must minimize the impact of using different chunk sizes.
What should you use? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Answer Area

Function:

- VECTOR_DISTANCE
- Vector_NORM
- VECTOR_SEARCH

Distance metric:

- cosine distance
- dot product
- Euclidean distance

Answer Area

Function: VECTOR_SEARCH

Distance metric: cosine distance

Q-You have a Microsoft SQL Server 2025 instance that has a managed identity enabled.
You have a database that contains a table named dbo.ManualChunks. dbo.ManualChunks contains product manuals.
A retrieval query already returns the top five matching chunks as nvarchar(max) text.
You need to call an Azure OpenAI REST endpoint for chat completions. The solution must provide the highest level of security.
You write the following Transact-SG1 code.

```
01 CREATE DATABASE SCOPED CREDENTIAL AzureOpenAIHeaders
02
03 GO
04 CREATE OR ALTER PROCEDURE dbo.AskManuals
05   ...
06   SELECT @chunks =
07   ...
08   SET @payload =
09   ...
10   EXEC @retval = sys.sp_invoke_external_rest_endpoint
11     @url = N'https://<your-resource>.openai.azure.com/openai/deployments/<your-deployment>/chat/completions?api-version=2024-02-15-preview',
12     @method = N'POST',
13     @credential = N'AzureOpenAIHeaders',
14     @payload = @payload,
15     @response = @response OUTPUT;
16 END;
17 GO
```

What should you insert at line 02?

A)
```
WITH IDENTITY = 'HTTPEndpointQueryString',
SECRET = N'{"api-key":"<value>"}';
```

B)
```
WITH IDENTITY = 'Managed Identity',
SECRET = '{"resourceid":"<value>"}';
```

C)
```
WITH IDENTITY = 'Managed Identity',
SECRET = N'{"api-key":"<value>"}';
```

D)
```
WITH IDENTITY = 'HTTPEndpointHeaders',
SECRET = N'{"api-key":"<value>"}';
```

E)
```
WITH IDENTITY = 'AzureOpenAI',
SECRET = N'{"resourceid":"<value>"}';
```

A. Option A
B. Option B
C. Option C
D. Option D
E. Option E

Answer- B

Q- You have an Azure SQL database.
You need to create a scalar user-defined function (UDF) that returns the number of whole years between an input parameter named 0orderDate and the current date/time as a single positive integer. The function must be created in Azure SQL Database. You write the following code.

```
01 CREATE FUNCTION dbo.ufnYearsSinceOrder (@OrderDate datetime2)
02 RETURNS int
03 AS
04 BEGIN
05
06 END
```

What should you insert at line 05? Options:
A. RETURN DATEDIFF(year, GETDATE(), @OrderDate);
B. DATEDIFF(month, @orderdate, GETDATE()) / 12
C. DATEPART(year, GETDATE()) - DATEPART(year, @orderdate)
D. **RETURN DATEDIFF(year, @OrderDate, GETDATE());**
Answer- D

Q- You have a GitHub Codespaces environment that has GitHub Copilot Chat installed and is connected to a SQL database in Microsoft Fabric named DB1 DB1 contains tables named Sales.Orders and Sales.Customers.
You use GitHub Copilot Chat in the context of DB1 .
A company policy prohibits sharing customer Personally Identifiable Information (Pll), secrets, and query result sets with any Al service.
You need to use GitHub Copilot Chat to write and review Transact-SQL code for a new stored procedure that will join Sales.Orders to sales .Customers and return customer names and email addresses. The solution must NOT share the actual data in the tables with GitHub Copilot Chat.
What should you do?
Options:
A. From Sales.Customers, paste several rows that include email addresses into a chat, so that GitHub Copilot Chat can infer edge cases.
B. Run a select statement that returns customer names and email addresses and provide the result set to GitHub Copilot Chat so that GitHub Copilot Chat can generate the stored procedure.
C. Provide the database connection string to GitHub Copilot Chat so that GitHub Copilot Chat can validate the stored procedure.
D. **Ask GitHub Copilot Chat to generate the stored procedure by using schema details only.**
Answer- D

Q- You have an Azure SQL table that contains the following data.

FaqId	ProductName	Question	Answer
1	Laptop Pro	Does it support USB-C charging?	Yes, it supports USB-C charging.
2	Tablet X	Does it support a stylus?	It supports the Active Pen stylus.

You need to retrieve data to be used as context for a large language model (LLM). The solution must minimize token usage.
Which formal should you use to send the data to the LLM?

```
[
  {
    "ProductName": "Laptop Pro",
    "Question": "Does it support USB-C charging?",
    "Answer": "Yes, it supports USB-C charging."
  },
  {
    "ProductName": "Tablet X",
    "Question": "Does it support a stylus?",
    "Answer": "It supports the Active Pen stylus."
  }
]
```

A)

```
[
  {
    "Prompt": "Yes, it supports USB-C charging."
  },
  {
    "Prompt": "It supports the Active Pen stylus."
  }
]
```

B)

```
[
  {


    "FaqId": 1,
    "Question": "Does it support USB-C charging?",
    "Answer": "Yes, it supports USB-C charging."
  },
  {
    "FaqId": 2,
    "Question": "Does it support a stylus?",
    "Answer": "It supports the Active Pen stylus."
  }
]
```

C)

```
[
  {
    "FaqId": 1,
    "ProductName": "Laptop Pro",
    "Question": "Does it support USB-C charging?",
    "Answer": "Yes, it supports USB-C charging."
  },
  {
    "FaqId": 2,
    "ProductName": "Tablet X",
    "Question": "Does it support a stylus?",
    "Answer": "It supports the Active Pen stylus."
  }
]
```

D)

Options:

A.**Option A**

B.Option B

C.Option C

D. Option D

Answer- A

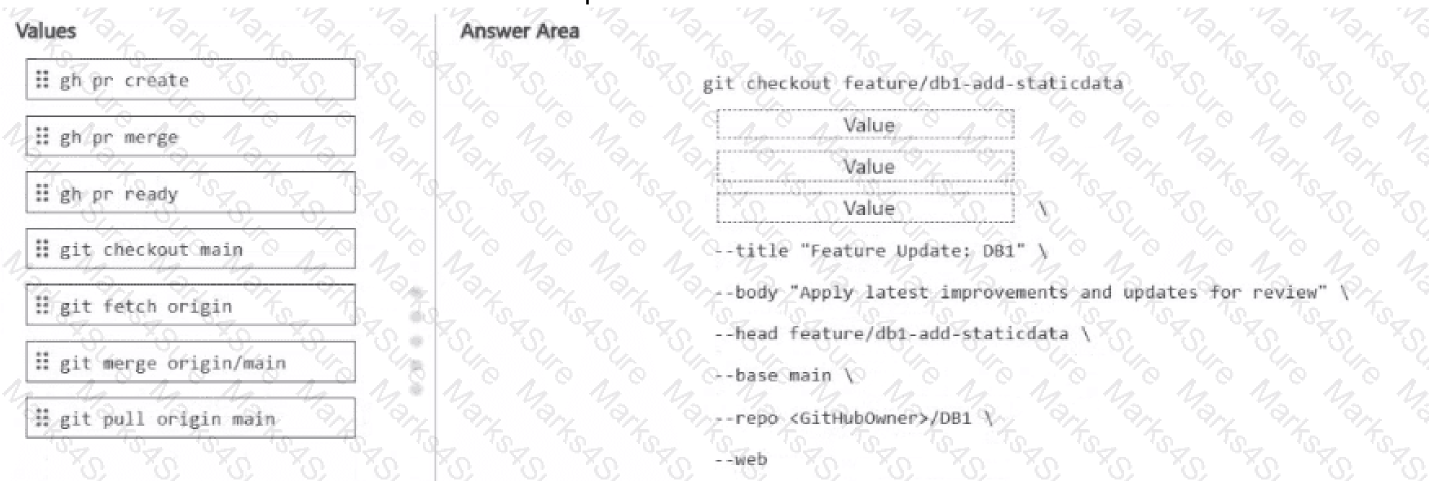
Q- You have a database named DB1. The schema is stored in a GitHub repository as an SDK style SQL database project.

You use a feature branch workflow to deploy changes to DB1

You need to update the local feature branch with the latest changes to main, and then create a pull request to merge the feature branch into main for review.

How should you complete the GitHub CLI script? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.



The correct sequence is:

git fetch origin

git merge origin/main

gh pr create

Q- You are creating a table that will store customer profiles.

You have the following Transact-SQL code.

```sql
CREATE TABLE dbo.CustomerProfiles
(
    CustomerId      BIGINT IDENTITY(1,1) PRIMARY KEY,
    FullName        NVARCHAR(200) MASKED WITH (FUNCTION = 'partial(1,"xxxx",1)'),
    EmailAddress    NVARCHAR(200) MASKED WITH (FUNCTION = 'email()'),
    PhoneNumber     NVARCHAR(50)  MASKED WITH (FUNCTION = 'default()'),
    RegionCode      NVARCHAR(10) NOT NULL
);
GO
CREATE FUNCTION dbo.fn_FilterByRegion(@RegionCode NVARCHAR(10))
RETURNS TABLE
AS
RETURN
(
     SELECT 1
     FROM dbo.UserRegionAccess ura
     WHERE ura.UserPrincipalName = SUSER_SNAME()
       AND ura.RegionCode = @RegionCode
);
GO
CREATE SECURITY POLICY CustomerRegionPolicy
ADD FILTER PREDICATE dbo.fn_FilterByRegion(RegionCode)
ON dbo.CustomerProfiles
WITH (STATE = ON);
GO
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No. NOTE: Each correct selection Is worth one point.

1. The schema meets the security requirements for PII data. → **Yes**
2. Administrators of the Azure SQL server can see all the rows in dbo.CustomerProfiles when they use an application. → **No**
3. The masking rules will apply even when row-level security (RLS) filters out rows. → **No**

Q- You have an Azure SQL database named SalesDB that contains tables named Sales.Orders and Sales.OrderLines. Both tables contain sales data

You have a Retrieval Augmented Generation (RAG) service that queries SalesDB to retrieve order details and passes the results to a large language model (ILM) as JSON text. The following is a sample of the JSON.

```json
[
  {
    "orderHeaderId": 102348,
    "orderNumber": "SO-2026-000912",
    "orderDateUtc": "2026-01-28T14:22:09Z",
    "customerId": 77821,
    "currencyCode": "EUR",
    "orderTotal": 149.97,
    "lines": [
      {
        "lineNumber": 1,
        "productId": 5012,
        "sku": "CBL-USB-C-1M",
        "quantity": 2,
        "unitPrice": 19.99,
        "lineTotal": 39.98
      },
      {
        "lineNumber": 2,
        "productId": 8841,
        "sku": "HUB-USB-C-7P",
        "quantity": 1,
        "unitPrice": 109.99,
        "lineTotal": 109.99
      }
    ]
  }
]
```

You need to return one 1SON document per order that includes the order header fields and an array of related order lines. The LIM must receive a single JSON array of orders, where each order contains a lines property that is a JSON array of line Items.

Which transact-SQL commands should you use to produce the required JSON shape from the relational tables? To answer, drag the appropriate commands to the correct operations. Each command may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content. NOTE: Each correct selection is worth one point.

Commands

- FOR JSON PATH
- JSON_MODIFY
- JSON_QUERY
- JSON_VALUE
- OPENJSON

Answer Area

Serialize the order-level JSON:

Generate a nested lines array:

Extract a single scalar value from the JSON text:

- Serialize the order-level JSON: FOR JSON PATH
- Generate a nested lines array: JSON_QUERY
- Extract a single scalar value from the JSON text: JSON_VALUE

Q- You have an Azure SQL database that has Query Store enabled

Query Performance Insight shows that one stored procedure has the longest runtime. The procedure runs the following parameterized query.

```
CREATE OR ALTER PROCEDURE dbo.GetRecentOrders
  @CustomerId int,
  @SinceDate  datetime2
AS
SELECT TOP (50)
    o.OrderId, o.OrderDate, o.Status, o.TotalAmount
FROM dbo.Orders AS o
WHERE o.CustomerId = @CustomerId
  AND o.OrderDate >= @SinceDate
ORDER BY o.OrderDate DESC;
```

The dbo.orders table has approximately 120 million rows. Customer-id is highly selective, and orderOate is used for range filtering and sorting.

You have the following indexes:

• Clustered index: PK_Orders on (Orderld)

• Nonclustered index: lx_0rders_order-Date on (OrderDate) with no included columns An actual execution plan captured from Query Store for slow runs shows the following:

• An index seek on ixordersorderDate followed by a Key Lookup (Clustered) on PKOrders for customerid, status, and TotalAnount

• A sort operator before top (50), because the results are ordered by orderDate DESC

For each of the following statements, select Yes if the statement is true. Otherwise, select No. NOTE:

Answer Area

Statements	Yes	No
To avoid the explicit sort for the query, create a nonclustered index on (CustomerId, OrderDate DESC) that includes (Status, TotalAmount).	◉	○
To eliminate the sort and make the query use an ordered seek, add CustomerId as an included column to IX_Orders_OrderDate.	○	◉
The plan indicates a bottleneck from a suboptimal query plan, rather locking or blocking.	◉	○

Q- You have an Azure SQL database named DB1 that contains two tables named knowledgebase and query_cache. knowledge_base contains support articles and embeddings. query_cache contains chat questions, responses, and embeddings DB1 supports an Al-enabled chat agent.

You need to design a solution that meets the following requirements:

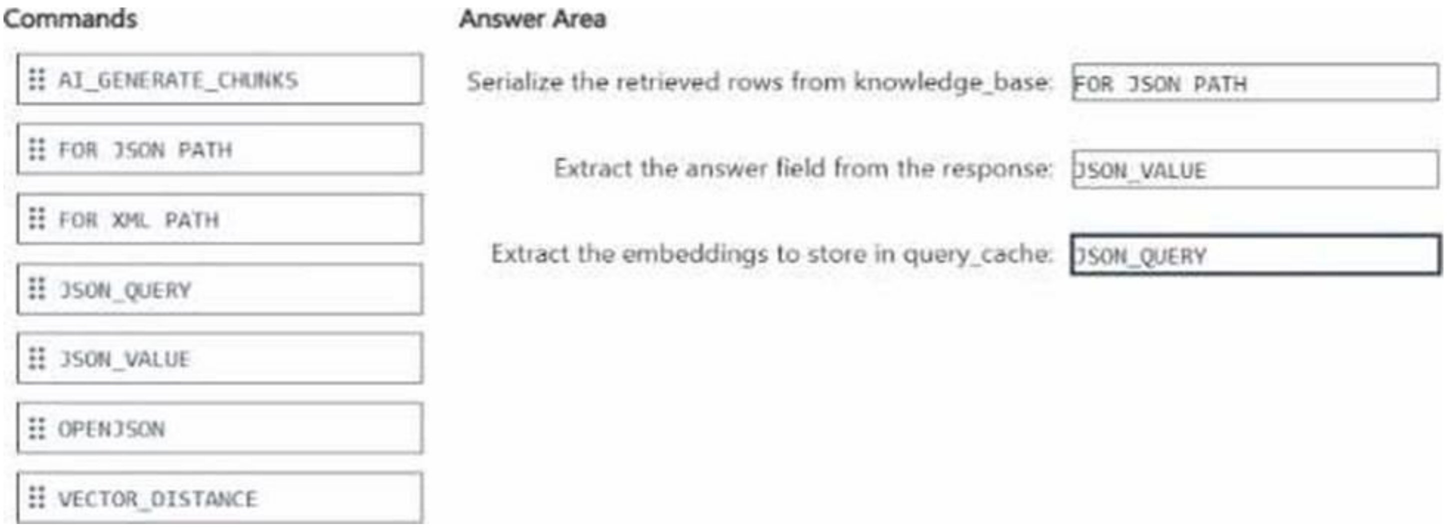
• Serializes the retrieved rows from knowledee_base

• Extracts the answer field from the response

• Extracts the embeddings to store in query_cache

You will call the external large language model (LLM) by using the sp_irwoke_external_re standpoint stored procedure.

Which Transact-SGL commands should you use for each requirement? To answer, drag the appropriate commands to the correct requirements. Each command may be used once, mote than once, or not at all. You may need to drag the split bar between panes or scroll to view content. NOTE: Each correct selection is worth one point.



The correct mapping is:
FOR JSON PATH
JSON_VALUE
JSON_QUERY

Q- You have an Azure SQL database that contains a table named dbo.Orders.
You have an application that calls a stored procedure named dbo.usp_tresteOrder to insert rows into dbo.Orders.
When an insert fails, the application receives inconsistent error details.
You need to implement error handling to ensure that any failures inside the procedure abort the transaction and return a consistent error to the caller.
How should you complete the stored procedure? To answer, drag the appropriate values to the correct targets, tach value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content. NOTE: Each correct selection is worth one point.

Values

- BEGIN CATCH
- IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION
- RAISERROR('CreateOrder failed', 16, 1)
- ROLLBACK TRANSACTION
- SET @OrderId = SCOPE_IDENTITY()
- THROW

Answer Area

```
CREATE OR ALTER PROCEDURE dbo.usp_CreateOrder
    @CustomerId int,
    @Amount decimal(10,2),
    @OrderId int OUTPUT
AS
BEGIN
    SET NOCOUNT ON;
    BEGIN TRY
        BEGIN TRANSACTION;
        INSERT INTO dbo.Orders(CustomerId, Amount, CreatedAt)
        VALUES (@CustomerId, @Amount, SYSUTCDATETIME());
        [ SET @OrderId = SCOPE_IDENTITY() ] ;
        COMMIT TRANSACTION;
    END TRY
    BEGIN CATCH
        [ IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION ] ;
        THROW;
    END CATCH
END
```

After the INSERT → SET @OrderId = SCOPE_IDENTITY()
Inside CATCH → IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION

The correct drag-and-drop choices are:
SET @OrderId = SCOPE_IDENTITY()
IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION

Q- You have a SQL database in Microsoft Fabric named Sales BD that contains a table named dbo.Products. You need to modify SalesBD to meet the following requirements:
*Create a vector index on the appropriate column.
* Use a supplied natural language query vector.
How should you complete the Transact-SQL code? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

The first correct selection is embedding because a vector index must be created on the vector column, not on a scalar distance column or a text column such as product_name. Microsoft's CREATE VECTOR INDEX documentation shows that the index is created directly on the vector-valued column, for example ON product_embeddings(embedding).

The second correct selection is VECTOR_SEARCH because the requirement is to use a supplied natural language query vector and search against the indexed embeddings. Microsoft documents that VECTOR SEARCH is the Transact-SQL function for approximate nearest neighbor vector retrieval and that it applies to SQL database in Microsoft Fabric as well as other supported SQL platforms.

This also matches the shown code pattern:

* declare a vector variable such as @query_vector VECTOR(1536),

*create a vector index on dbo.Products(embedding),

query with VECTOR_SEARCH(... SIMILAR TO = @query_vector, METRIC = 'cosine ', TOP_N= 10).

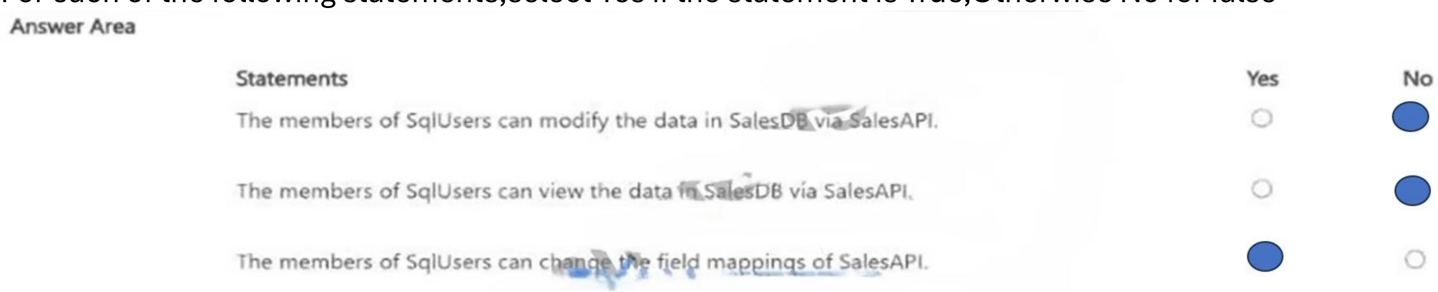
Q- You have a Microsoft Fabric workspace named Workspace1 that contains a SQL database named SalesDB and an API for GraphQL tern named SalesApi.

You have a Microsoft Entra group named SqlUsers.

From Workspace1, you assign permission to SalesApi as shown in the following exhibit.

The connection to SalesDB has the connectivity option configured as shown in the above exhibit(right):

SqlUser has the Viewer role for Workspace1
For each of the following statements,select Yes if the statement is True,Otherwise No for false



Q- You need to recommend a solution to lesolve the slow dashboard query issue. What should you recommend?
A. On Fleetid, create a filtered index where lastupdatedutc > DATEADD(DAV, -7, SYSUTCOATETIME()).
B. On Fleetid, create a nonclustered index that includes Lastupdatedutc. inginestatus, and BatteryHealth.
C. On Lastupdatedutc. create a nonclustered index that includes Fleetid.
D. Create a clustered index on Lastupdatedutc.

Answer- B

Q- You have an Azure subscription. The subscription contains an Azure SQL database named SalesDB and an Azure App Service app named sales-api. sales-api uses virtual network integration to a subnet named vnet- prod/subnet-app and reads from SalesDB You need to configure authentication and network access to meet the following requirements:
* Ensure that sales-api connects to SalesDB by using passwordless authentication.
* Ensure that all the database traffic remains within the subscription.
The solution must minimize administrative effort
com

What should you configure? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Q- You have a GitHub Codespaces environment that has GitHub Copilot Chat installed and is connected to a SQL database in Microsoft Fabric named DB1 DB1 contains tables named Sales. Orders and Sales. Customers. You use GitHub Copilot Chat in the context of DB1.

A company policy prohibits sharing customer Personally Identifiable Information (PII), secrets, and query result sets with any Al service.

You need to use GitHub Copilot Chat to write and review Transact-SQL code for a new stored procedure that will join Sales. Orders to sales .Customers and return customer names and email addresses. The solution must NOT share the actual data in the tables with GitHub Copilot Chat.

What should you do?

A. Run a select statement that returns customer names and email addresses and provide the result set to GitHub Copilot Chat so that GitHub Copilot Chat can generate the stored procedure.

B. Ask GitHub Copilot Chat to generate the stored procedure by using schema details only.

C. Provide the database connection string to GitHub Copilot Chat so that GitHub Copilot Chat can validate the stored procedure.

D. From Sales.Customers, paste several rows that include email addresses into a chat, so that GitHub Copilot Chat can infer edge cases.

Answer- B

Updated Questions

Question 1: You have an Azure SQL database containing a table named Rooms. The Rooms table was created by using the following Transact-SQL statement:

CREATE TABLE Rooms

)

RoomID int PRIMARY KEY,

Owner nvarchar(100), Capacity int

);
You find that some records in the Rooms table have NULL values in the Owner field. You need to ensure that every future record includes a value for the Owner field. What should you add?

A. a foreign key

B. a check constraint

C. a nonclustered index

D. a unique constraint

Correct answer:
B. **a check constraint**
Here's why:
• The requirement is to prevent NULL values in the Owner column for future inserts/updates. • In SQL Server, this is enforced by either:
• Declaring the column as NOT NULL at creation, or
• Adding a CHECK constraint that disallows NULL.
• Foreign key (A) ensures referential integrity, not non-null enforcement.
• Nonclustered index (C) improves query performance, does not enforce data integrity.
· Unique constraint (D) → ensures uniqueness, but still allows NULL unless explicitly disallowed.

Question 2: DRAG DROP-
You have an Azure SQL database that contains a table named dbo.Orders.
You have an application that calls a stored procedure named dbo.usp_CreateOrder to insert rows into dbo.Orders.
When an insert fails, the application receives inconsistent error details.
You need to implement error handling to ensure that any failures inside the procedure abort the transaction and return a consistent error to the caller.
How should you complete the stored procedure? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Answer Area

```
CREATE OR ALTER PROCEDURE dbo.usp_CreateOrder
    @CustomerId int,
    @Amount decimal(10,2),
    @OrderId int OUTPUT
AS
BEGIN
    SET NOCOUNT ON;
    BEGIN TRY
        BEGIN TRANSACTION;
        INSERT INTO dbo.Orders(CustomerId, Amount, CreatedAt)
        VALUES (@CustomerId, @Amount, SYSUTCDATETIME());
        SET @OrderId = SCOPE_IDENTITY() ;
        COMMIT TRANSACTION;
    END TRY
    BEGIN CATCH
        IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION
        THROW;
    END CATCH
END
```

Question 3: Your team is developing an Azure SQL dataset solution from a locally cloned GitHub repository by using Microsoft Visual Studio Code and GitHub Copilot Chat.
You need to disable the GitHub Copilot repository-level instructions for yourself without affecting other users. What should you do?

A. From Visual Studio Code, modify your GitHub Copilot Chat user settings.

B. Add a --debug flag when you start the GitHub Copilot Chat extension.
C. Delete .github/copilot-instructions.md.

Answer -A

Question 4: You have a NODE table named dbo.Person and an EDGE table named dbo.Knows. Each row

in dbo.Person contains: • PersonID (int)
•DisplayName (nvarchar(100))
You need to use the MATCH operator and exactly two directed Knows relationships to return the PersonID and DisplayName of people reachable from a person identified by an input parameter @StartPersonId.
Which Transact-SQL query should you use?

Correct Answer: D

Question 4: You have a NODE table named dbo.Person and an EDGE table named dbo.Knows.Each row in dbo.Person contains:
- PersonID (int)
- DisplayName (nvarchar(100))

You need to use the MATCH operator and exactly two directed Knows relationships to return the PersonID and DisplayName of people reachable from a person identified by an input parameter @StartPersonId.
Which Transact-SQL query should you use?

A. SELECT p2.PersonId, @StartPersonId
FROM dbo.Person AS p1, dbo.Knows AS k1, dbo.Person AS p2, dbo.Knows AS k2, dbo.Person AS p3
WHERE p1.DisplayName = p2.DisplayName
AND MATCH(p1-(k1)->p2-(k2)->p3);

B. SELECT p3.PersonId, p3.DisplayName FROM dbo.Person AS p1
JOIN dbo.Knows AS k1 ON 1 = 1
JOIN dbo.Person AS p2 ON 1 = 1
JOIN dbo.Knows AS k2 ON 1 = 1
JOIN dbo.Person AS p3 ON 1 = 1
WHERE p1.PersonId = @StartPersonId
AND MATCH(p3<-(k2)-p2<-(k1)-p1);

C. SELECT p3.PersonId, p3.DisplayName
FROM dbo.Person AS p1, dbo.Knows AS k1, dbo.Person AS p2, dbo.Knows AS k2, dbo.Person AS p3
WHERE p1.PersonId = @StartPersonId
AND MATCH(p1-(k1)->p2) AND MATCH(p2-(k2)->p3);

D. SELECT p3.PersonId, p3.DisplayName
FROM dbo.Person AS p1, dbo.Knows AS k1, dbo.Person AS p2, dbo.Knows AS k2, dbo.Person AS p3
WHERE p1.PersonId = @StartPersonId
AND MATCH(p1-(k1)->p2-(k2)->p3);

Correct Answer: D

Question 5: You have a SQL database that contains a column named Payload, which stores customer data in JSON format. Example JSON:
{

"date": "2020-01-25",
"customer_email": "user@contoso.com"
}
Some customers use subaddressing in their email addresses, such as: user1+promo@contoso.com You need to normalize these emails by removing the subaddressing so that: user+promo@contoso.com →user1@contoso.com
Which Transact-SQL expression should you use?
A. REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+.*$', '')
B.REGEXP_SUBSTR(JSON_VALUE(Payload, '$.customer_email'), '^[^+]+@.*$=')
C.REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), \+*@', '@')
D.REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+*', "')

Answer-C

Question 5: You have a SQL database that contains a column named Payload, which stores customer data in JSON format.
Example JSON:

```
{
  "date": "2020-01-25",
  "customer_email": "user@contoso.com"
}
```

Some customers use subaddressing in their email addresses, such as: user1+promo@contoso.com
You need to normalize these emails by removing the subaddressing so that: user1+promo@contoso.com → user1@contoso.com
Which Transact-SQL expression should you use?

A. REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+.*$', '')
B. REGEXP_SUBSTR(JSON_VALUE(Payload, '$.customer_email'), '^[^+]+@.*$=')
C. REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+.*@', '@')
D. REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+.*', '')

Correct answer:
C. REGEXP_REPLACE(JSON_VALUE(Payload, '$.customer_email'), '\+.*@', '@')

Question 6: Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution. After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.
You have an SDK-style SQL database project stored in a Git repository. The project targets an Azure SQL database.
The CI build fails with unresolved reference errors when the project leferences system objects. You need to update the SQL database project to ensure that dotnet build the correct system objects in the database model for Azure SQL Database.
Solution: Add an artifact reference to the Azure SQL Database maste validates successfully by including
Does this meet the goal?
A. Yes
B. No
Correct Answer: B. No

Question 7: You have an Azure SQL database. You need to create a scalar user-defined function (UDF) that returns the number of whole years between an input parameter named @OrderDate and the current date/time as a single positive integer. The function must be created in an Azure SQL Database.
You write the following code:
01 CREATE FUNCTION dbo.ufnYearsSinceOrder (@OrderDate datetime)
02 RETURNS int
03 AS
04 BEGIN
05
06 END

What should you insert at line 05?

A. RETURN DATEDIFF(year, GETDATE(), @OrderDate);

B. DATEDIFF(month, @orderdate, GETDATE()) /12

C. DATEPART(year, GETDATE()) - DATEPART(year, @orderdate)

D. **RETURN DATEDIFF(year, @OrderDate, GETDATE());**

Question 7: You have an Azure SQL database. You need to create a scalar user-defined function (UDF) that returns the number of whole years between an input parameter named @OrderDate and the current date/time as a single positive integer. The function must be created in an Azure SQL Database.

You write the following code:

```
01 CREATE FUNCTION dbo.ufnYearsSinceOrder (@OrderDate datetime)
02 RETURNS int
03 AS
04 BEGIN
05
06 END
```

What should you insert at line 05?

A. RETURN DATEDIFF(year, GETDATE(), @OrderDate);
B. DATEDIFF(month, @orderdate, GETDATE()) / 12
C. DATEPART(year, GETDATE()) - DATEPART(year, @orderdate)
D. RETURN DATEDIFF(year, @OrderDate, GETDATE());

Correct answer:
D. RETURN DATEDIFF(year, @OrderDate, GETDATE());

Question 8: You have an Azure SQL database.

You deploy Data API builder (DAB) to Azure Container Apps by using the mcr.microsoft.com/azure-databases/data- api-builder:latest image. You have the following Container Apps secrets:

• MSSQL_CONNECTION_STRING that maps to the SQL connection string

• DAB_CONFIG_BASE64 that maps to the DAB configuration

You need to initialize the DAB configuration to read the SQL connection stri

Which command should you run?

Question 8: You have an Azure SQL database.
You deploy Data API builder (DAB) to Azure Container Apps by using the mcr.microsoft.com/azure-databases/data-api-builder:latest image.
You have the following Container Apps secrets:

- MSSQL_CONNECTION_STRING that maps to the SQL connection string
- DAB_CONFIG_BASE64 that maps to the DAB configuration

You need to initialize the DAB configuration to read the SQL connection string.
Which command should you run?

A. dab init --database-type mssql --connection-string "secretref:DAB_CONFIG_BASE64" --host-mode Production --config dab-config.json
B. dab init --database-type mssql --connection-string "@env('MSSQL_CONNECTION_STRING')" --host-mode Production --config dab-config.json
C. dab init --database-type mssql --connection-string "secretref:mssql-connection-string" --host-mode Production --config dab-config.json
D. dab init --database-type mssql --connection-string "@env('DAB_CONFIG_BASE64')" --host-mode Production --config dab-config.json

Correct answer:
B. dab init --database-type mssql --connection-string "@env('MSSQL_CONNECTION_STRING')" --host-mode Production --config dab-config.json

Question 9: You have a SQL database in Microsoft Fabric that contains a nvarchar (max) column named MessageText. An ID is always contained within the first paragraph of MessageText.
You need to write a Transact-SQL query that uses REGEXP_SUBSTR to extract the ID from MessageText.
What should you include in the query?

Question 9: You have a SQL database in Microsoft Fabric that contains a nvarchar (max) column named MessageText. An ID is always contained within the first paragraph of MessageText.
You need to write a Transact-SQL query that uses REGEXP_SUBSTR to extract the ID from MessageText.
What should you include in the query?

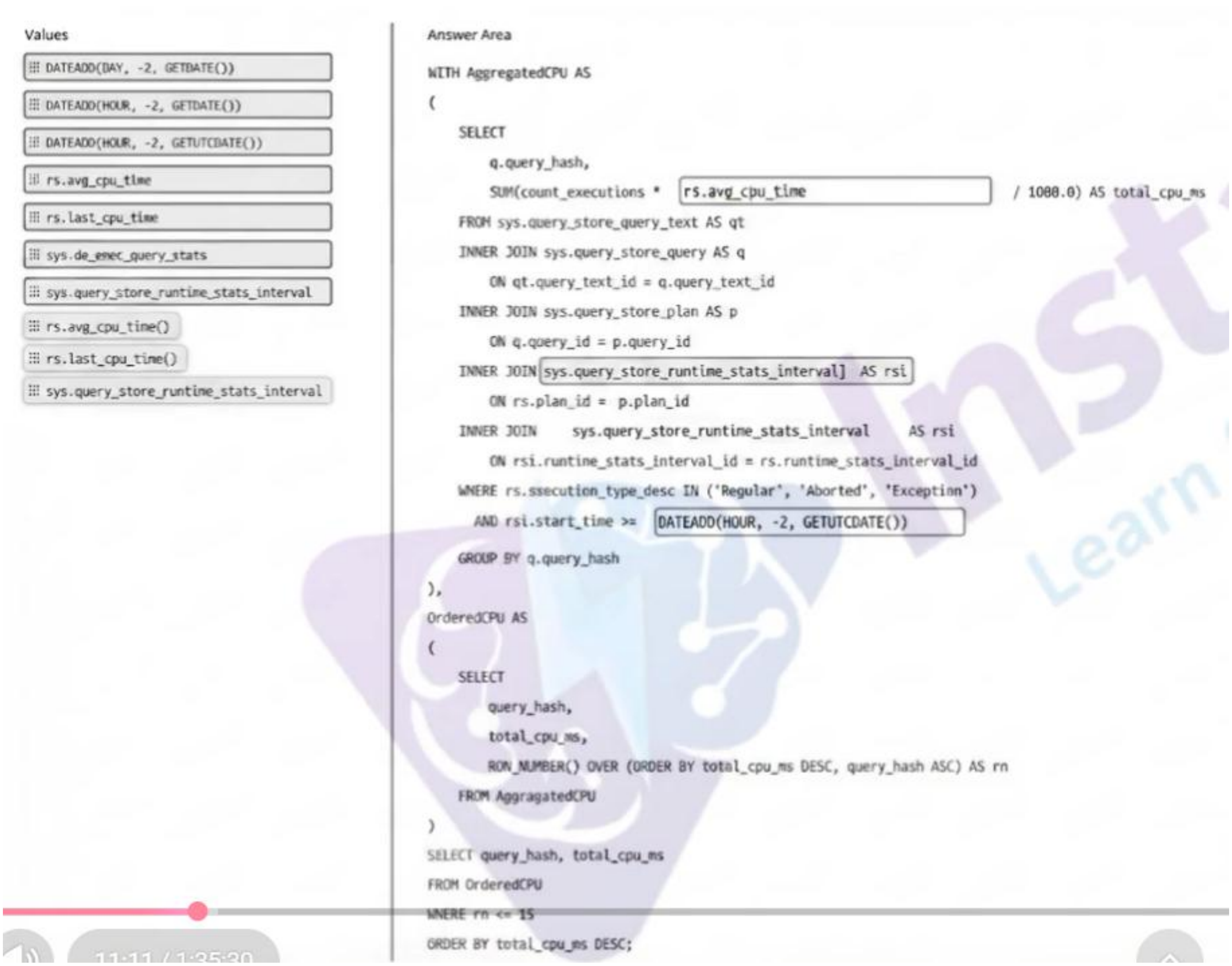
A. Apply STRING_ESCAPE(MessageText, 'json') before calling REGEXP_SUBSTR.
B. Cast MessageText to nvarchar (4000) before calling REGEXP_SUBSTR.
C. Add a COLLATE Latin1_General_CS_AS clause to MessageText before calling REGEXP_SUBSTR.
D. Run TRY_CONVERT(varchar(max), MessageText) before calling REGEXP_SUBSTR.

Correct answer:
B. Cast MessageText to nvarchar(4000) before calling REGEXP_SUBSTR.

- This ensures the regex function can process the string.
- It's explicitly documented in DP-800 prep materials as the required step when working with nvarchar(max) columns.

Question 10: You have an Azure SQL database that contains database-level Data Definition Language (DDL) triggers, including a trigger named ddl_Audit.
You need to prevent ddl_Audit from firing during the next deployment. The trigger object must remain in place. Which Transact-SQL statement should you use?
A. ALTER TRIGGER
B. ALTER DATABASE
C. ALTER SERVER AUDIT SPECIFICATION
D. **DISABLE TRIGGER**
E. ALTER DATABASE AUDIT SPECIFICATION
Correct answer: D. DISABLE TRIGGER

Question 11: Drag and Drop Question
You have an Azure SQL database that supports an AI-driven product search API.
You need to identify the top CPU-consuming queries from the last two hours by using Query Store data. The solution must aggregate CPU consumption across executions and return only the top 15 query hashes.
How should you complete the Transact-SQL code? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. NOTE: Each correct selection is worth one point.



Question 12: You need to design a generative AI solution that uses a Microsoft SQL Server 2025 database named DB1 as a data source.

The solution must generate responses that meet the following requirements:

Responses are grounded in the latest transactional and reference data stored in DB1.

Responses must not require retraining or fine-tuning the language model when the data changes.

Responses can include citations or references to the source data used.

Which scenario is the best use case for implementing a Retrieval Augmented Generation (RAG) pattern? (More than one answer choice may achieve the goal. Select the BEST answer.)

A. Summarizing free-form user input text

B. Training a custom language model on historical database data

C. Answering user questions based on company-specific knowledge

D. Generating marketing slogans based on user sentiment analysis

Correct Answer: C. Answering user questions based on company-specific knowledge

Question 13: You have an Azure SQL database that contains tables named dbo.ProductDocs and dbo.ProductDocsEmbeddings. dbo.ProductDocs contains product documentation with columns:

DocId (int)
Title (nvarchar(200))
Body (nvarchar(max)) LastModified (datetime2)
The documentation is edited throughout the day.
dbo.ProductDocsEmbeddings contains:
DocId (int)
ChunkOrder (int) ChunkText (nvarchar(max)) Embedding (vector(1536))

The current embedding pipeline runs once per night. You need to ensure that embeddings are updated every time the The solution must not require a nightly batch process. What should you include in the solution?
A. Fixed-size chunking
B. A smaller embedding model
C. Table triggers
D. **Change tracking on dbo.ProductDocs**

Correct Answer: D. Change tracking on dbo.ProductDocs

Question 13: You have an Azure SQL database that contains tables named dbo.ProductDocs and dbo.ProductDocsEmbeddings.

dbo.ProductDocs contains product documentation with columns:
DocId (int)
Title (nvarchar(200))
Body (nvarchar(max))
LastModified (datetime2)
The documentation is edited throughout the day.

dbo.ProductDocsEmbeddings contains:
DocId (int)
ChunkOrder (int)
ChunkText (nvarchar(max))
Embedding (vector(1536))

The current embedding pipeline runs once per night.
You need to ensure that embeddings are updated every time the underlying documentation content changes.
The solution must not require a nightly batch process.
What should you include in the solution?

A. Fixed-size chunking
B. A smaller embedding model
C. Table triggers
D. Change tracking on dbo.ProductDocs

Correct Answer:
D. Change tracking on dbo.ProductDocs

Question 14: You have a GitHub Actions workflow that builds and deploys an Azure SQL database. The schema is stored in a GitHub repository as an SDK-style SQL database project.
Following a code review, you discover that you need to generate a report that shows whether the production schema has diverged from the model in source control.
Which action should you add to the pipeline?

A. **SqlPackage.exe /Action:DriftReport**

B.SqlPackage.exe /Action:Deploy Report
C. SqlPackage.exe /Action:Extract
D.SqlPackage.exe /Action:Script
Correct Answer: A. SqlPackage.exe /Action:DriftReport.

Question 15: You have an Azure SQL database that includes a column named Notes.
A security review finds that Notes contains sensitive data.
You need to protect the data so that neither stored values nor query inputs disclose information about the actual data.
The solution must prevent a user from inferring data relationships or repetitions from the encrypted output. Which should you use?
A. Always Encrypted with secure enclaves
B. **Always Encrypted with randomized encryption**
C. Row-level security (RLS)
D. Always Encrypted with deterministic encryption

Correct Answer: B. Always Encrypted with randomized encryption

Question 16: You have a SQL database in Microsoft Fabric that includes the following functions:
• A multi-statement table-valued function (TVF) named Sales.mstvf_OrderStatus() that returns order status information.
• A scalar user-defined function (UDF) named dbo.ufn_GetTaxMultiplier(@TaxAmt money, @StateCode char(2)) that returns a numeric multiplier used in tax calculations.
Reporting queries frequently join Sales.mstvf_OrderStatus() to Sales.SalesOrderHeader and return large result sets. A performance review shows that the queries generate inconsistent execution plans.
During a code review, a developer finds that the following Transact-SQL statement produced an error:
EXEC @ret = ufn_GetTaxMultiplier @TaxAmt = 100.00, @StateCode = 'WA';
For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Statement	Yes	No
You can use `GETDATE()` in `dbo.ufn_GetTaxMultiplier` to produce nondeterministic results.	Yes	
Rewriting `Sales.mstvf_OrderStatus()` as an inline table TVF will reduce the number of inconsistent execution plans.	Yes	
Replacing `ufn_GetTaxMultiplier` with `dbo.ufn_GetTaxMultiplier` in the `EXEC` statement will resolve the error.		No

Question 17: You have an Azure SQL database that contains order data.
A reporting query that aggregates monthly revenue for each customer runs frequently. You need to reduce the time required to retrieve the computed values.
The solution must not change any underlying table structure.
What should you do?
A. Create a view by using ORDER BY without TOP, and then create a unique clustered index on the view.

B. Create a view without using WITH SCHEMABINDING, and then create a nonclustered index on the view.
C. Create a view by using GROUP BY, and then create a unique clustered index on the view.
D. **Create a view by using WITH SCHEMABINDING, include COUNT_BIG(*), and then create a unique clustered index on the view.**

Correct Answer: D. Create a view by using WITH SCHEMABINDING, include COUNT BIG(*)....

Question 18: You have a Microsoft SQL Server 2025 database containing a table named dbo.CustomerMessages.
• dbo.CustomerMessages has two columns:
• MessageID (int)
• MessageRaw (nvarchar(max))

MessageRaw can include a phone number in several formats, and some rows
You need to write one SELECT query that meets these requirements:
• Return MessageID, RawNumber, DigitsOnly, and PhoneStatus.
include a phone number.
• RawNumber must contain the first substring that matches a phone-number pattern, or NULL when no match is found.
• DigitsOnly must remove every non-digit character from RawNumber, or return NULL.
Missinstatus must return Valid when MessageRaw contains a phone number; otherwise, it must return
Each value may be used once, more than once, or not at all.

```
SELECT
    MessageID,
    REGEXP_SUBSTR  MessageRaw, '\d{3}[\.\-\s]\d{3}[\.\-\s]\d{4}') AS RawNumber,
    REGEXP_REPLACE (REGEXP_SUBSTR(MessageRaw, '\d{3}[\.\s]\d{4}'), '\D', '') AS DigitsOnly,
    CASE
        WHEN  REGEXP_LIKE  MessageRaw, '\d{3}[\.\-\s]\d{3}[\.\-\s]\d{4}') = 1
        THEN 'Valid'
        ELSE 'Missing'
    END AS PhoneStatus
FROM dbo.CustomerMessages;
```

Requirement	Function	Reason
Return the first matching phone number	`REGEXP_SUBSTR`	Returns the first substring matching the regex or `NULL` if no match exists.
Remove non-digit characters	`REGEXP_REPLACE(..., '\D', '')`	Replaces every non-digit with an empty string. If `REGEXP_SUBSTR` returns `NULL`, the result remains `NULL`.
Determine whether a phone number exists	`REGEXP_LIKE`	Returns `1` when the pattern matches and `0` otherwise.

Question 19: You have an Azure SQL database named SalesDB that includes a table named dbo.Articles. dbo.Articles holds two million articles with embeddings.
The articles are updated frequently during the day.
You query the embeddings by using VECTOR_SEARCH.
Users report that semantic search results do not reflect updates until the next day.
You need to ensure the embeddings are updated whenever the articles change. The solution must minimize CPU usage on SalesDB.
Which embedding-maintenance method should you implement?

A. Modify the query to use VECTOR_DISTANCE instead of VECTOR_SEARCH.
B.Enable change data capture (CDC) on dbo.Atricles and use an Azure function sapp to process CDC changes.
C. Run an hourly Transact-SQL job that regenerates embeddings for all the rows in dbo.Articles.
D. On dbo.Articles, create a trigger that calls AI_GENERATE_EMBEDDINGS for each inserted or updated row.
Answer- B

Question 20: You have an Azure SQL database containing a table named dbo.Orders. dbo.Orders has a CreateDate column that stores order-creation dates. You need to create a stored procedure that filters Orders by CreateDate for one calendar day. The solution must be SARGable (able to use indexes efficiently).How should you complete the Transact-SQL code?

```
CREATE PROCEDURE dbo.usp_SearchOrders
    @StartDate date
AS
BEGIN
    SET NOCOUNT ON;

    DECLARE @EndDate date;

    SET @EndDate = [ DATEADD(day, 1, @StartDate) ];

    SELECT o.CreateDate,
           o.OrderId,
           o.ShipDate
    FROM dbo.Orders AS o
    WHERE o.CreateDate >= [ @StartDate ]
      AND o.CreateDate < [ @EndDate ];
END;
GO
```

Question 21: You have an Azure SQL database named SalesDB that includes a table named dbo.Products.

- dbo.Products has three columns: Embedding, Category, and Price.
  The Embedding column is defined as VECTOR(1536).
- You use AI_GENERATE_EMBEDDINGS and VECTOR_SEARCH to enable semantic search and apply additional filters to Category and Price.
- You plan to change the embedding model from text-embedding-ada-002 to text-embedding-3-small.
- Existing rows already have embeddings in the Embedding column.

You need to implement the model change. Applications must be able to use VECTOR_SEARCH without runtime errors.

What should you do first?

A. **Regenerate embeddings for the existing rows.**

B. Normalize the vector lengths before storing new embeddings.

C. Convert the Embedding column to nvarchar(max).

D. Create a vector index on dbo.Products.Embedding.

Correct Answer: A. Regenerate embeddings for the existing rows

Question 22: DRAG & DROP-

You have an Azure SQL database that supports an OLTP application.

You need to write Transact-SQL code that returns blocking-chain details. The output must return only sessions that are blocked or that are blocking other sessions.

Complete the code.

WITH cteBL (session_id, blocking these) AS

Each value may be used once, more than once, or not at all.

```sql
WITH cteBL (session_id, blocking_these) AS
(
    SELECT
        s.session_id,
        blocking_these = x.blocking_these
    FROM sys.dm_exec_sessions AS s
    CROSS APPLY
    (
        SELECT
            ISNULL(CONVERT(varchar(6), er.session_id), '') + ',' + ' '
        FROM sys.dm_exec_requests AS er
        WHERE er.blocking_session_id = ISNULL(s.session_id, 0)
            AND er.blocking_session_id <> 0
        FOR XML PATH('')
    ) AS x(blocking_these)
)
SELECT
    s.session_id,
    blocked_by = r.blocking_session_id,
    bl.blocking_these,
    batch_text = t.text,
    input_buffer = ib.event_info
FROM sys.dm_exec_sessions AS s

LEFT OUTER JOIN sys.dm_exec_requests AS r

ON r.session_id = s.session_id

INNER JOIN cteBL AS bl
    ON s.session_id = bl.session_id

OUTER APPLY sys.dm_exec_sql_text(r.sql_handle) AS t

OUTER APPLY sys.dm_exec_input_buffer(s.session_id, NULL) AS ib

WHERE bl.blocking_these IS NOT NULL
    OR r.blocking_session_id > 0
ORDER BY LEN(bl.blocking_these) DESC,
         r.blocking_session_id DESC,
         r.session_id;
```

Blank	Value
First blank	**LEFT OUTER JOIN sys.dm_exec_requests AS r**
Second blank	**OUTER APPLY sys.dm_exec_sql_text(r.sql_handle) AS t**
Third blank	**OUTER APPLY sys.dm_exec_input_buffer(s.session_id, NULL) AS ib**

Case Study 1:
This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam. You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.
To answer the questions included in a case study, you will need to reference information that is provided in the case study. Case studies might contain exhibits and other resources that provide more information about the scenario that is described in the case study. Each question is independent dent of the other questions in this case study. At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.
case
To start this case study,To display the 1st question click the Next button u
Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.
Existing Environment
Azure Environment
Contoso has an Azure subscription in North Europe that contains the corporate infrastructure. The current infrastructure contains a Microsoft SQL Server 2017 database. The database contains the following tables.

Question 23: HOTSPOT -

Case Study 1:
This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam. You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.
To answer the questions included in a case study, you will need to reference information that is provided in the case study. Case studies might contain exhibits and other resources that provide more information about the scenario that is described in the case study. Each question is independent of the other questions in this case study.
At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.
To start the case study
To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment and problem statements. If the case study has an All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.
Existing Environment
Azure Environment
Contoso has an Azure subscription in North Europe that contains the corporate infrastructure. The current infrastructure contains a Microsoft SQL Server 2017 database. The database contains the following tables.

Case Study 1:

Table	Columns
CustomerFeedback	FeedbackId *(int, PK)* FeedbackJson *(nvarchar(max))*
Fleets	FleetId *(int, PK)* FleetName *(nvarchar(100))* Description *(nvarchar(256))*
MaintenanceEvents	MaintenanceId *(int, PK)* VehicleId *(int)* LastModifiedUTC *(datetime2)* Description *(nvarchar(256))*
SupportTickets	TicketId *(int, PK)* FleetId *(int)* CreatedUtc *(datetime2)*
UserAccounts	UserId *(int, PK)* UserPrincipalName *(nvarchar(256))* JobRole *(nvarchar(256))* StartDate *(datetime2)*
VehicleIncidentReports	IncidentId *(int, PK)* VehicleId *(int)* FleetId *(int)* IncidentType *(nvarchar(50))* VehicleLocation *(nvarchar(200))* IncidentDescription *(nvarchar(max))* SeverityScore *(int)*
Vehicles	VehicleId *(int, PK)* VIN *(nvarchar(50))* VehicleDescription *(nvarchar(256))*
VehicleHealthSummary	VehicleId *(int, PK)* FleetId *(int)* Summary *(nvarchar(2000))* LastUpdatedUtc *(datetime2)* EngineStatus *(bit)* EngineStatusLastUpdatedUtc *(datetime2)* BatteryHealth *(int)* Embeddings *(vector(1536))*

The FeedbackJson column has a full-text index and stores JSON documents in the following format.

```
{
   "text": "The battery drains too fast when driving uphill.",
   "category": "Battery",
   "metadata": {
      "appVersion": "5.2.1",
      "device": "Android",
      "language": "en-GB"
   }
}
```

The support staff at Contoso never has the UNMASK permission.
Problem Statements
Contoso is deploying a new Azure SQL database that will become the authoritative data store for the following:
AI workloads
Vector search
Modernized API access
Retrieval Augmented Generation (RAG) pipelines
Sometimes the ingestion pipeline fails due to malformed JSON and duplicate payloads.
The engineers at Contoso report that the following dashboard query runs slowly.

```
SELECT VehicleId, LastUpdatedUtc, EngineStatus, BatteryHealth
FROM dbo.VehicleHealthSummary
WHERE FleetId = @FleetId
ORDER BY LastUpdatedUtc DESC;
```

Case Study 1:

You review the execution plan and discover that the plan shows a clustered index scan.
VehicleIncidentReports often contains details about the weather, traffic conditions, and location. Analysts report that it is difficult to find similar incidents based on these details.
Requirements
Planned Changes
Contoso wants to modernize Fleet Intelligence Platform to support AI-powered semantic search over incident reports.
Security Requirements
Contoso identifies the following security requirements:
Restrict the support staff from viewing Personally Identifiable Information (PII) data, which is full email addresses and phone numbers.
Enforce row-level filtering so that analysts see only incidents for the fleets to which they are assigned. The analysts can be assigned to multiple fleets.
Database Performance and Requirements
Contoso identifies the following telemetry requirements:
Telemetry data must be stored in a partitioned table.
Telemetry data must provide predictable performance for ingestion and retention operations.
latitude, longitude, and accuracy JSON properties must be filtered by using an index seek.
Contoso identifies the following maintenance data requirements:
Ensure that any changes to a row in the MaintenanceEvents table updates the corresponding value in the LastModifiedUtc column to the time of the change.
Avoid recursive updates.
AI Search, Embeddings, and Vector Indexing
Contoso plans to implement semantic search over incident data to meet the following requirements:
Embeddings must be stored in dedicated Azure SQL Database tables.
Embeddings must be generated from rich natural language fields.
Chunking must preserve semantic coherence.

Case Study 1:

Hybrid search must combine the following:
Vector similarity
Keyword filtering or boosting
Development Requirements
The development team at Contoso will use Microsoft Visual Studio Code and GitHub Copilot and will retrieve live metadata from the databases.
Contoso identifies the following requirements for querying data in the FeedbackJson column of the CustomerFeedback table:
Extract the customer feedback text from the JSON document.
Filter rows where the JSON text contains a keyword.
Calculate a fuzzy similarity score between the feedback text and a known issue description.
Order the results by similarity score, with the highest score first.

You need to meet the development requirements for the FeedbackJson column.
How should you complete the Transact-SQL query?

To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Hybrid search must combine the following:
Vector similarity
Keyword filtering or boosting
Development Requirements
The development team at Contoso will use Microsoft Visual Studio Code and GitHub Copilot and will retrieve live metadata from the databases.
Contoso identifies the following requirements for querying data in the FeedbackJson column of the CustomerFeedback table:
Extract the customer feedback text from the JSON document.
Filter rows where the JSON text contains a keyword.
Calculate a fuzzy similarity score between the feedback text and a known issue description. Order the results by similarity score, with the highest score first.
You need to meet the development requirements for the FeedbackJson column. How should you complete the Transact-SQL query?
To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
SELECT
  f.FeedbackId,
  f.VehicleId,
```

- CONTAINS(FeedbackJson, @Keyword)
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.details.comment'), @Keyword) < 3
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.text'), @Keyword) < 3
- JSON_QUERY(f.FeedbackJson, '$.text', @KnownIssueDescription) AS FeedbackText
- JSON_VALUE(f.FeedbackJson, '$.text') AS FeedbackText
- SimilarityScore

,

```
  EDIT_DISTANCE_SIMILARITY(
    JSON_VALUE(f.FeedbackJson, '$.text'),
    @KnownIssueDescription
  ) AS SimilarityScore
FROM
  dbo.CustomerFeedback f
WHERE
```

- CONTAINS(FeedbackJson, @Keyword)
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.details.comment'), @Keyword) < 3
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.text'), @Keyword) < 3
- JSON_QUERY(f.FeedbackJson, '$.text', @KnownIssueDescription) AS FeedbackText
- JSON_VALUE(f.FeedbackJson, '$.text') AS FeedbackText
- SimilarityScore

```
ORDER BY
```

- CONTAINS(FeedbackJson, @Keyword)
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.details.comment'), @Keyword) < 3
- EDIT_DISTANCE(JSON_VALUE(f.FeedbackJson, '$.text'), @Keyword) < 3
- JSON_QUERY(f.FeedbackJson, '$.text', @KnownIssueDescription) AS FeedbackText
- JSON_VALUE(f.FeedbackJson, '$.text') AS FeedbackText
- SimilarityScore

```
DESC;
```

Question 25: You need to satisfy the database-performance requirements for maintenance data.
Complete the Transact-SQL code.
Each value may be used once, more than once, or not at all.

Correct answers:

- inserted contains the rows affected by the UPDATE statement.
- m.MaintenanceId = i.MaintenanceId joins the updated rows back to the original table using the primary key.
- i.MaintenanceId IS NOT NULL ensures only valid updated rows are processed.
- SYSUTCDATETIME() stores the audit timestamp in UTC.
- This is a set-based trigger pattern recommended for SQL Server/Azure SQL performance.

```
CREATE TRIGGER dbo.trgMaintenanceEvents_UpdateTimestamp
    ON dbo.MaintenanceEvents
    AFTER UPDATE
AS
BEGIN
    UPDATE m
    SET LastModifiedUtc = SYSUTCDATETIME()
    FROM dbo.MaintenanceEvents m
    INNER JOIN inserted i
        ON
            m.MaintenanceId = i.MaintenanceId

    WHERE
            i.MaintenanceId IS NOT NULL
                                            ;
END;
GO
```

Case Study 1:

Question 24: You need to recommend a solution that resolves the ingestion-pipeline failure issues.
Which two actions should you recommend? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.
Choose two
A. Enable snapshot isolation on the database.
B. Use a trigger to automatically rewrite malformed JSON. C. Add foreign key constraints on the table.
D. Create a unique index on a hash of the payload.
E. Add a check constraint that validates the JSON structure.
Answer: D,E

Question 26: You need to recommend a solution that resolves the slow dashboard-query issue. What should you recommend?
A. Create a clustered index on LastUpdated Utc.
B. On FleetId, create a nonclustered index that includes LastUpdated Utc, EngineStatus, and BatteryHealth.
C. On LastUpdated Utc, create a nonclustered index that includes FleetId.
D. On FleetId, create a filtered index where LastUpdated Utc > DATEADD(DAY, -7, SYSUTCDATETIME()).
Answer-B

Question 27: You need to create a table in the database to store telemetry data. You have the following Transact-SQL code:

```sql
CREATE TABLE dbo.VehicleTelemetry
(
    TelemetryId BIGINT IDENTITY(1,1) NOT NULL,
    VehicleId NVARCHAR(50) NOT NULL,
    TelemetryTimeUtc DATETIME2(3) NOT NULL,
    BatteryPercent TINYINT NULL,
    SpeedKmh SMALLINT NULL,
    LocationJson JSON NULL,
    ErrorCodesJson JSON NULL,
    RawPayload NVARCHAR(MAX) NULL,
    SysStartTime DATETIME2(7) GENERATED ALWAYS AS ROW START NOT NULL,
    SysEndTime DATETIME2(7) GENERATED ALWAYS AS ROW END NOT NULL,
    PERIOD FOR SYSTEM_TIME (SysStartTime, SysEndTime),
    CONSTRAINT PK_VehicleTelemetry PRIMARY KEY CLUSTERED (TelemetryId)
)
WITH
(
    SYSTEM_VERSIONING = ON
    (
        HISTORY_TABLE = dbo.VehicleTelemetryHistory
    )
);
GO

CREATE INDEX IX_VehicleTelemetry_Time
    ON dbo.VehicleTelemetry (TelemetryTimeUtc);

CREATE JSON INDEX JI_VehicleTelemetry_Location
    ON dbo.VehicleTelemetry (LocationJson) FOR
(
    '$.location.latitude',
    '$.location.longitude',
    '$.location.accuracy'
);
```

Statement	Answer	Explanation
The code meets the database performance requirements for partitioning.	No	The code does not create a partition function or partition scheme. Creating indexes or a temporal history table does not enable partitioning.
The code meets the database performance requirements for JSON property querying.	Yes	A JSON index is created on `LocationJson` for `latitude`, `longitude`, and `accuracy`, improving queries that filter these JSON properties.
Queries that filter on `$.location.heading` will use the `JI_VehicleTelemetry_Location` index.	No	The JSON index does not include `$.location.heading`; it only supports the JSON paths defined in the index.

Question 28: You need to create embeddings to address the issues reported by the analysts. Which column should you use?

A. VehicleLocation

B. IncidentDescription

C. IncidentType

D. SeverityScore

Answer -B

Question 29: Contoso wants to modernize the Fleet Intelligence Platform to support Al-powered semantic search over incident reports and health summary data.

Requirement:

You need to enable similarity search so analysts can retrieve the most relevant summary reports. The solution must minimize latency.

What should you include in the solution?

A. A computed column that manually compares vector values

B. A standard non-clustered index on the Embeddings (vector(1536)) column East.

C. A full-text index on the Embeddings (vector(1536)) column

D. A vector index on the Embeddings (vector(1536)) column.

Q- You are creating a table that will store customer profiles.You have the following Transact-SQL code:

Statement	Answer	Explanation
The schema meets the security requirements for PII data.	Yes	Dynamic Data Masking (DDM) protects sensitive columns, and Row-Level Security (RLS) restricts users to only the rows for their assigned region.
Administrators of the Azure SQL server can see all the rows in `dbo.CustomerProfiles` when they use an application.	No	RLS applies to application queries, including administrators, unless they use a principal that explicitly bypasses the security policy.
The masking rules will apply even when row-level security (RLS) filters out rows.	No	RLS filters rows first. Rows that are filtered out are never returned, so Dynamic Data Masking is not applied to them.

```sql
CREATE TABLE dbo.CustomerProfiles
(
    CustomerId BIGINT IDENTITY(1,1) PRIMARY KEY,
    FullName NVARCHAR(200) MASKED WITH (FUNCTION = 'partial(1,"xxxx",1)')
    EmailAddress NVARCHAR(200) MASKED WITH (FUNCTION = 'email()'),
    PhoneNumber NVARCHAR(50) MASKED WITH (FUNCTION = 'default()'),
    RegionCode NVARCHAR(10) NOT NULL
);
GO

CREATE FUNCTION dbo.fn_FilterByRegion(@RegionCode NVARCHAR(10))
RETURNS TABLE
AS
RETURN
(
    SELECT 1
    FROM dbo.UserRegionAccess ura
    WHERE ura.UserPrincipalName = SUSER_SNAME()
        AND ura.RegionCode = @RegionCode
);
GO

CREATE SECURITY POLICY CustomerRegionPolicy
ADD FILTER PREDICATE dbo.fn_FilterByRegion(RegionCode)
ON dbo.CustomerProfiles
WITH (STATE = ON);
GO
```

Question 31: You need to recommend a solution that enables the development team to retrieve live metadata and meets the development requirements.
What should you include in the recommendation?
A. Export the database schema as a .dacpac file and load the schema into a GitHub Copilot context window.
B. Add the schema to a GitHub Copilot instruction file.
C. Use an MCP server.
D. Include the database project in the code repository.
Answer-C

Question 32: You have a GitHub Codespaces environment with GitHub Copilot Chat installed that is

connected to a Microsoft Fabric SQL database named DB1. DB1 contains tables named Sales. Orders and Sales.Customers.
You use GitHub Copilot Chat in the context of DB1.
A company policy prohibits sharing customer Personally Identifiable Information (PII), secrets, and query result sets with any AI service.
You need to use GitHub Copilot Chat to write and review Transact-SQL code for a new stored procedure that joins Sales. Orders to Sales.Customers and returns customer names and email addresses .The solution must NOT share the actual data in the tables with GitHub Copilot Chat.
What should you do?

A. From Sales.Customers, paste several rows that include email addresses into a chat, so that GitHub Copilot Chat can infer edge cases.
B. Run a SELECT statement that returns customer names and email addresses and provide the result set to GitHub Copilot Chat so that Copilot can generate the stored procedure.
C. Provide the database connection string to GitHub Copilot Chat so that Copilot can validate the stored procedure.
D. Ask GitHub Copilot Chat to generate the stored procedure by using schema details only.
Answer-D

Question 33: Your company has an ecommerce catalog in a Microsoft SQL Server 2025 database called SalesDB. SalesDB includes a table named products. The products table has these columns:
• product_id (int)
• product_name (nvarchar(200))
description (nvarchar(max))
• category (nvarchar(50))
• brand (nvarchar(50))
• price (decimal)
•sku (nvarchar(40))

The description fields are updated each day, and price may change several times daily. You want customers to submit natural-language queries and use structured filters for brand and price.
You plan to store embeddings in a new VECTOR(1536) column and use VECTOR_SEARCH(... METRIC='cosine' ...). For each statement, select Yes if it is true. Otherwise, select No

Correct answers:

Statements	Answer	Explanation
Generating an embedding by concatenating `product_name`, `category`, and `description` will support the customer requirements.	Yes	These columns contain semantic text information. Combining them creates an embedding that can support natural-language similarity search. Brand and price can still be applied as structured SQL filters.
Including `price` in the text used to generate embeddings is required.	No	Price is a structured numeric value that changes frequently. It should be filtered using SQL predicates (for example, `WHERE price BETWEEN ...`) rather than included in embeddings.
The underlying base type of the embeddings will be `float(32)`.	Yes	SQL Server vector columns use 32-bit floating-point values (`float(32)`) to store embedding dimensions such as `VECTOR(1536)`.

Question 34: You have a SQL database in Microsoft Fabric that includes a table named WebSite.Logs storing application telemetry data.
The table has an nvarchar(max) column named log containing JSON documents.
A daily report:

• Filters on the $.severity JSON property.
• Returns:

-LogId
-LogDateTime

- log

Currently, the query performs full table scans You need to modify WebSite.Logs so that:

• Filtering by $.severity is efficient.
• Key lookups are avoided. Complete the Transact-SQL code.

```
ALTER TABLE WebSite.Logs
ADD severity
 AS JSON_VALUE([log], '$.severity') PERSISTED
;
GO


CREATE INDEX ix_severity
ON WebSite.Logs(severity)
 INCLUDE (LogId, LogDateTime, [log])
;
GO
```

Item	Answer	Explanation
Computed column definition	✅ `AS JSON_VALUE([log], '$.severity') PERSISTED`	`JSON_VALUE()` extracts a scalar JSON value. A persisted computed column stores the value physically, allowing SQL Server to index it efficiently.
Index definition	✅ `INCLUDE (LogId, LogDateTime, [log])`	The query returns these columns. Including them makes the index covering and avoids key lookups to the base table.
Why not `JSON_QUERY()`	❌	`JSON_QUERY()` returns JSON objects/arrays. `severity` is a scalar property, so `JSON_VALUE()` is required.
Why not only `INCLUDE(log)`	❌	The report also returns `LogId` and `LogDateTime`; excluding them would still require lookups.

Question 35: You have an Azure SQL database with Query Store enabled. Query Performance Insight identifies one stored procedure as having the longest runtime. The procedure executes the following parameterized query:
The dbo.Orders table contains approximately 120 million rows.
- Customerld is highly selective.
- OrderDate is used for range filtering and sorting.

You have these indexes:
-Clustered index: PK_Orders on (Orderld)
- Nonclustered index: IX_Orders_OrderDate on (OrderDate) with no included columns.
The actual execution plan from Query Store for slow executions shows:
•An index seek on IX_Orders_OrderDate, followed by a Key Lookup (Clustered) on PK_Orders for Customerld, Status, and TotalAmount.
• A Sort operator before TOP (50) because results are ordered by OrderDate DESC.

```
CREATE OR ALTER PROCEDURE dbo.GetRecentOrders
    @CustomerId int,
    @sinceDate datetime2
AS
SELECT TOP (50)
    o.OrderId,
    o.OrderDate,
    o.Status,
    o.TotalAmount
FROM dbo.Orders AS o
WHERE o.CustomerId = @CustomerId
    AND o.OrderDate >= @sinceDate
ORDER BY o.OrderDate DESC;
```

Statements	Answer	Explanation
To avoid the explicit sort for the query, create a nonclustered index on `(CustomerId, OrderDate DESC)` that includes `(Status, TotalAmount)`.	Yes	This index matches the filtering and ordering pattern. SQL Server can seek by `CustomerId`, retrieve rows already sorted by `OrderDate DESC`, and avoid the Sort operator. Included columns remove Key Lookups.
To eliminate the sort and make the query use an ordered seek, add `CustomerId` as an included column to `IX_Orders_OrderDate`.	No	Included columns are only stored for covering queries. They do not define index order. The index is still ordered only by `OrderDate`, so it cannot efficiently seek by `CustomerId` and avoid sorting.
The plan indicates a bottleneck from a suboptimal query plan, rather than locking or blocking.	Yes	Query Store shows expensive operators (Key Lookup and Sort). There is no evidence of blocking waits or lock contention. The issue is indexing/query optimization.

Question 36: You have an Azure SQL database named SalesDB that contains tables named Sales. Orders and Sales.OrderLines. Both tables hold sales data.
A Retrieval Augmented Generation (RAG) service queries SalesDB to retrieve order details and passes the results to a large language model (LLM) as JSON text. The following is a simple example of the JSON.

You need to return one JSON document per order that includes the order header fields and an array of related order lines. The LLM must receive a single JSON array of orders, where each order contains a lines property that is a JSON array of line items. Which Transact-SQL commands should you use to produce the required JSON shape from the relational tables?
Each command may be used once, more than once, or not at all

ask	Answer
enerate a nested lines array	
erialize the order-level JSON	
xtract a single scalar value from JSON text	

Drag Options

- `FOR JSON PATH`
- `JSON_MODIFY`
- `JSON_QUERY`
- `JSON_VALUE`
- `OPENJSON`

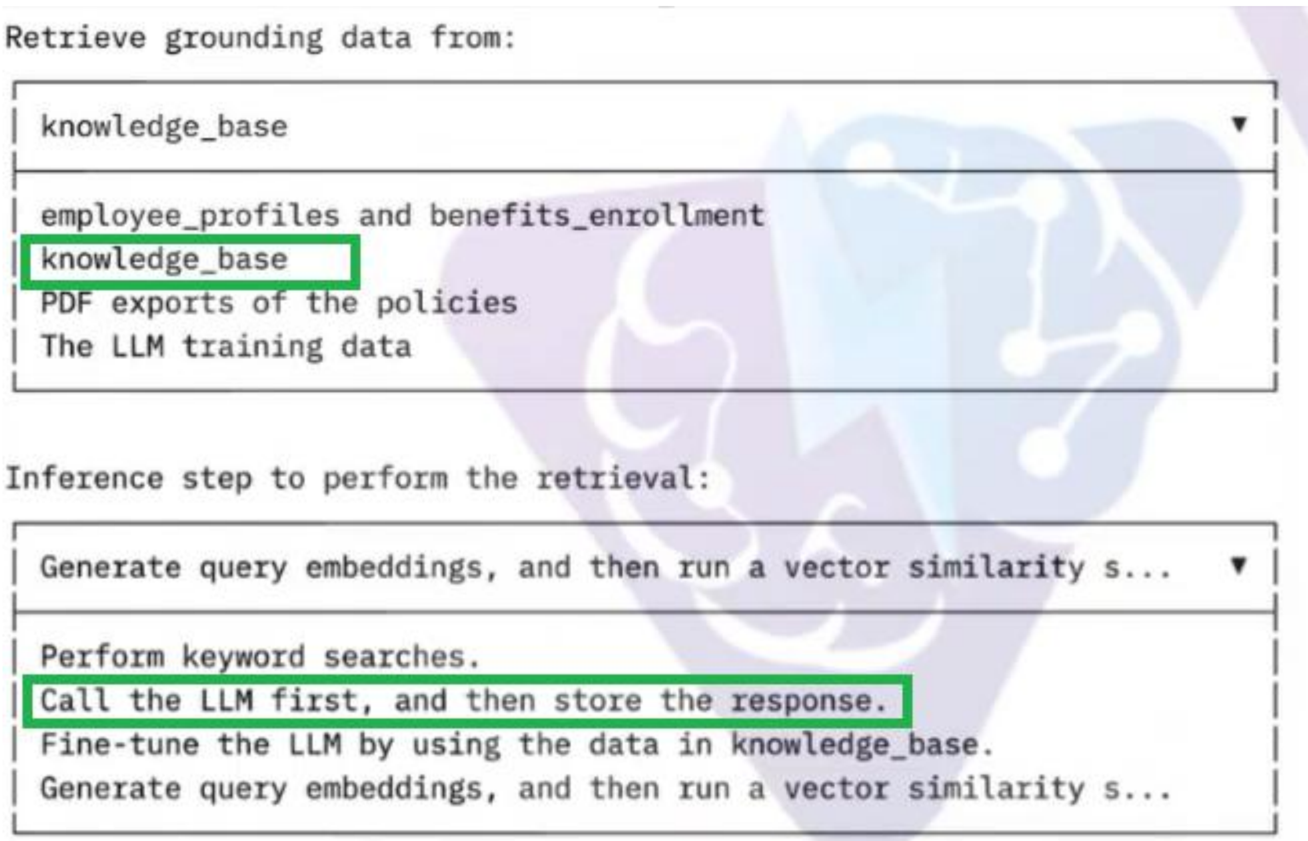
Task	Correct Answer	Why it is Correct
Generate a nested lines array	**FOR JSON PATH**	`FOR JSON PATH` converts the related `Sales.OrderLines` rows into a JSON array, which becomes the `lines` property for each order.
Serialize the order-level JSON	**FOR JSON PATH**	The outer `FOR JSON PATH` serializes all orders into a single JSON array, which is the format required by the RAG service and LLM.
Extract a single scalar value from JSON text	**JSON_VALUE**	`JSON_VALUE` extracts a single scalar value (such as a string, number, or date) from a JSON document. It is intended for scalar values, unlike `JSON_QUERY`, which returns objects or arrays.

JSON

```json
[
  {
    "orderHeaderId": 102348,
    "orderNumber": "SO-2026-000912",
    "orderDateUtc": "2026-01-28T14:22:09Z",
    "customerId": 77821,
    "currencyCode": "EUR",
    "orderTotal": 149.97,
    "lines": [
      {
        "lineNumber": 1,
        "productId": 5012,
        "sku": "CBL-USB-C-1M",
        "quantity": 2,
        "unitPrice": 19.99,
        "lineTotal": 39.98
      },
      {
        "lineNumber": 2,
        "productId": 8841,
        "sku": "HUB-USB-C-7P",
        "quantity": 1,
        "unitPrice": 109.99,
        "lineTotal": 109.99
      }
    ]
  }
]
```

Question 37: Hotspot -
You have an Azure SQL database that contains a table named knowledge_base. knowledge_base stores human resources (HR) policy documents and contains columns named title, content, category, and

embedding.
You have an application named App1. App1 queries two relational tables named employee_profiles and benefits_enrollment that contain HR data. App1 hosts a chatbot that calls a large language model (LLM) directly.
Users report that the chatbot answers general HR questions correctly but provides outdated or incorrect answers when policies change. The chatbot also fails to answer questions that reference internal policy documents by title or category.
You need to recommend a Retrieval Augmented Generation (RAG) solution resolve the chatbot issues.
What should you recommend? To answer, select the appropriate Options to in the answer area.NOTE: Each correct selection is worth one point.



Question 38: You have a Microsoft SQL Server 2025 instance that has a managed identity enabled. You have a database that contains a table named dbo.ManualChunks.

- dbo.ManualChunks contains product manuals.
- A retrieval query already returns the top five matching chunks as nvarchar(max) text.

You need to call an Azure OpenAI REST endpoint for chat completions. The solution must provide the highest level of security. You write the following Transact-SQL code:

```
01 CREATE DATABASE SCOPED CREDENTIAL AzureOpenAIHeaders
02 ___________
03 GO
04 CREATE OR ALTER PROCEDURE dbo.AskManuals
05 AS
06 SELECT @chunks = ...
07 SET @payload = ...
08 EXEC @retval = sys.sp_invoke_external_rest_endpoint
09    @url = N'https://<your-resource>.openai.azure.com/openai/deployments/<your-deployment>/chat/completions?api-version=2024-02-15-preview',
10    @method = N'POST',
11    @credential = N'AzureOpenAIHeaders',
12    @payload = @payload,
13    @response = @response OUTPUT;
14 END;
15 GO
```


What should you insert at line 02?

What should you insert at line 02?
A. WITH IDENTITY = 'HTTPEndpointQueryString',
SECRET = N'("api-key":"<value>")';

B. WITH IDENTITY = 'Managed Identity',
SECRET = N'("resourceid":"<value>")';

C. WITH IDENTITY = 'Managed Identity',
SECRET = N'("api-key":"<value>")';

D. WITH IDENTITY = 'HTTPEndpointHeaders',
SECRET = N'("api-key":"<value>")';

E. WITH IDENTITY = 'Managed Identity';

Answer- B

Correct Answer:
B. WITH IDENTITY = 'Managed Identity', SECRET = '{"resourceid":"https://cognitiveservices.azure.com"}';

Question 39: You have an Azure SQL database that contains a table named dbo.ManualChunks.
* dbo.Manual Chunks contains product manuals.
• A retrieval query already returns the top five matching chunks as nvarchar(max) text.
You need to call an Azure OpenAI REST endpoint for chat completions. The request body must include both the user question and the retrieved chunks.
You write the following Transact-SQL code: What should you insert at line **22**?
A. FOR XML AUTO, TYPE, XML SCHEMA
B. FOR JSON AUTO, INCLUDE_NULL_VALUES
C. FOR XML PATH, INCLUDE NULL VALUES
D. **FOR JSON PATH, WITHOUT_ARRAY_WRAPPER**
Answer-D

```
01 CREATE DATABASE SCOPED CREDENTIAL AzureOpenAIHeaders
02 WITH IDENTITY = 'HTTPEndpointHeaders',
03 SECRET = N'("api-key": "<YOUR_AZURE_OPENAI_API_KEY>")';
04 GO
05 CREATE OR ALTER PROCEDURE dbo.AskManuals
06 AS
07 BEGIN
08   DECLARE @chunks = (
09     SELECT TOP (5)
10       mc.ChunkText AS [text]
11     FROM dbo.ManualChunks AS mc
12     ORDER BY mc.Score DESC
13     FOR JSON PATH
14   );
15   SET @payload = (
16     SELECT
17       'system' AS [messages[0].role],
18       'Use only the provided manual chunks.' AS [messages[0].content],
19       'user' AS [messages[1].role],
20       CONCAT(@question, CHAR(10), JSON_QUERY(@chunks)) AS [messages[1].content]
21       -- Line 22
22
23     );
24   EXEC @retval = ...
25 END;
26 GO
```

Question 40: You have an Azure SQL database named ToDo that contains a table named dbo.ToDo. Your company plans to develop an Azure Functions app to run whenever the rows in dbo.ToDo change. The app will process INSERT, UPDATE, and DELETE events by using the Azure SQL trigger binding. You need to configure ToDo to support the planned app.

What should you do?

A. Enable change tracking on ToDo and dbo.ToDo.

B. On dbo.ToDo, create a Data Manipulation Language (DML) trigger that calls Azure Functions HTTP endpoint.

C. Enable change data capture (CDC) on ToDo and dbo.ToDo.

D. On dbo.ToDo, create a Data Definition Language (DDL) trigger that calls the Azure Functions HTTP endpoint.

Question 41: Which Azure service is used for visualizing SQL + AI insights?

A. Power BI

B. Azure CLI

C. Azure DNS

D. Azure Firewall

Answer- A

Question 42: What is Retrieval-Augmented Generation (RAG)?
A. A backup strategy
B. Combining LLMs with external data sources
C. A SQL indexing method
D. A query optimization technique

Question 43: Your development team uses Microsoft Visual Studio Code with the MSSQL extension and the GitHub Copilot Chat extension.
• The team connects to an Azure SQL database by using individual database logins.
• They use the @mssql chat participant to generate and run Transact-SQL queries from prompts.
What is used to ensure that GitHub Copilot Chat-generated queries run in the context of the developer?
A. SQL permissions
B. Azure role-based access control (Azure RBAC) permissions
C. GitHub organization permissions
D. Shared MCP instruction files

Question 44: Hotspot -
You have an Azure Al Search service and an index named hotels that includes a vector field named DescriptionVector.
You query hotels by using the Search Documents REST API.
You add semantic ranking to the hybrid search query and discover:
• Some queries return fewer results than expected.
• Captions and answers are missing.
You need to complete the hybrid search request to meet the following requirements:
• Include more documents when ranking.
• Always include captions and answers.

Correct answers:

- queryType = "semantic" → ensures semantic ranking is applied.
- captions = "extractive" → guarantees captions are always included.
- answers = "extractive" → guarantees answers are always included.
- k = 50 and maxTextRecallSize = 50 → expand recall, ensuring more documents are considered during ranking (solves the "fewer results than expected" issue).
- semanticConfiguration = "hotels" → applies the semantic configuration defined for the index.

```
{
  "search": "ocean view",
  "vectorqueries": [
    {
      "vector": [/* embedding array */],
      "fields": "DescriptionVector",
      "k": 50
      "kind": "vector"
    }
  ],
  "queryType": "semantic"
  "semanticConfiguration": "hotels",
  "captions": "extractive"
  "answers": "extractive"
  "top": 10,
  "hybridSearch": { "maxTextRecallSize": 50 }
}
```

Question 45: Why is indexing important for vector search?
A. Reduces storage
B. Improves similarity search performance
C. Encrypts data
D. Enables backups
Answer- B

Question 46: What is the primary purpose of Azure AI in SQL development?
A. Replace SQL Server
B. Automate infrastructure deployment
C. Enhance querying and analytics with AI capabilities
D. Eliminate the need for databases

Q:HOTSPOT:

You have an Azure SQL database that contains the following tables and columns:

Tables	Columns
Articles	• ArticleId (int) • Title (nvarchar(200)) • Notes (nvarchar(max)) • Description (nvarchar(max))
NotesEmbeddings	• ArticleId (int) • ChunkIndex (int) • Embedding (vector(1536))
DescriptionEmbeddings	• ArticleId (int) • ChunkIndex (int) • Embedding (vector(1536))

Embeddings in the NotesEmbeddings and DescriptionEmbeddings tables have been generated from values in the Description and Notes columns of the Articles table by using different chunk sizes.
You need to perform approximate nearest neighbor (ANN) queries across both embedding tables. The solution must minimize the impact of using different chunk sizes.What should you use? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Function:
- VECTOR_DISTANCE
- VECTOR_NORM
- VECTOR_SEARCH

Distance metric:
- cosine distance
- dot product
- Euclidean distance

Here's why:

- VECTOR_SEARCH is the correct function for ANN queries in Azure SQL over vector columns.
- Cosine distance normalizes vector magnitudes, which minimizes the impact of different chunk sizes (since chunk size affects embedding length/magnitude).
- Euclidean distance and dot product are magnitude-sensitive, so they would be skewed by chunk size differences.

Question 48: Hotspot -
You have a SQL database that contains a table named Sales.Orders. An application uses two stored procedures to access the table.
During monitoring, several sessions are found suspended with: wait_type = LCK_M_X
All sessions share the same:
- wait_resource mapped to Sales.Orders
- nonzero blocking_session_id
The sys.dm_exec_input_buffer for the blocking session returns a last submitted command of :

BEGIN TRANSACTIONUPDATE Sales.Orders

The sys.dm_exec_sessions shows the blocking session as:
- sleeping
- open_transaction_count = 1

Statement	Correct Answer	Explanation
The blocking is caused by an uncommitted explicit transaction in the blocking session that is holding locks.	Yes	The blocking session executed `BEGIN TRANSACTION` followed by `UPDATE Sales.Orders`. The session is now sleeping but still has `open_transaction_count = 1`, which means the transaction is not completed. The UPDATE operation holds exclusive locks on the affected rows/pages/table. Because the transaction remains open, the locks are not released and other sessions can be blocked.
While an UPDATE operation on Sales.Orders is occurring, SELECT statements will be blocked.	No	This statement is too broad. A normal SELECT under the default READ COMMITTED isolation level is not always blocked by an UPDATE, especially when row versioning is enabled in Azure SQL Database. SELECT statements are blocked only when they require incompatible locks or use locking hints such as `UPDLOCK`, `HOLDLOCK`, or higher isolation levels.
Joining `sys.dm_tran_locks` to `sys.dm_exec_requests` will show which session holds locks involved in the blocking chain.	Yes	`sys.dm_tran_locks` provides information about locks, resources, lock modes, and owning sessions. `sys.dm_exec_requests` provides active request information including `session_id` and `blocking_session_id`. Joining these views helps identify the blocking session and the locks causing the wait.

Question 49: You have an Azure SQL database that contains tables named dbo.Tickets and dbo.TicketNotes.

*dbo.Tickets contains support tickets.

• dbo.TicketNotes contains ticket notes.

A retrieval query returns the top five relevant ticket notes for a user question.

You plan to implement a Retrieval Augmented Generation (RAG) pattern that meets the following requirements:

- Formats the retrieved relational data for large language model (LLM) processing.
- Sends the user question and retrieved context to an Azure OpenAI REST endpoint for chat completions
- Extracts the response text from the LLM respontWhich Transact-SQL function should you use to extract the response text?

A. JSON_MODIFY(@response, '$.result')
B. OPENJSON(@response)
C. JSON_QUERY(@response, '$.choices')
D. JSON_VALUE(@response, '$.choices[0].message.content')
Answer- D

Question 51: You have a Microsoft SQL Server 2025 instance that contains a database named SalesDB. SalesDB supports a Retrieval Augmented Generation (RAG) pattern for internal support tickets. The SQL Server instance runs without any outbound network connectivity.
You plan to generate embeddings inside the SQL Server instance and store them in a table for vector similarity queries.
You need to ensure that only a database user account named AlApplicationUser can run embedding generation by using the model. Which two actions should you perform?
Each correct answer presents part of the solution. NOTE: Each correct selection is worth one point..

A. Grant the control permission on SalesDB to AlApplicationUser.

B. Create a database audit specification on SalesDB owned by AlApplicationUser.
C. Grant the execute permission on the external model project to AlApplicationUser.
D. Create an external model project by using ONNX runtime and local paths.
E. Create an external model project that points to a Microsoft Foundry REST endpoint.
Answer-C,D

Question 52: What is a common pattern for integrating LLMs with SQL?
A. Direct SQL execution inside LLM
B. RAG pipeline
C. Manual CSV export
D. Backup restore
Answer- B

Question 54: You have an Azure SQL database with table Customer:
• NationalIDNumber→ unique per customer
• InvestigationNotes free-form text
Stored procedure does point lookups by NationalIDNumber and returns Investigation Notes. You need to encrypt both columns using Always Encrypted with the highest security possible.

A. deterministic for National IDNumber and randomized for Investigation Notes
B. deterministic for National IDNumber and deterministic for Investigation Notes
C. randomized for NationalIDNumber and randomized for Investigation Notes
D. randomized for NationalIDNumber and deterministic for Investigation Notes
Answer-A

Question 55: You have an Azure SQL database named ProductsDB. A Data API Builder (DAB) is deployed to Azure Container Apps using the image: mcr.microsoft.com/azure-databases/data-api-builder:latest.
The container app has the following configuration:
• Secrets:
  -mssql-connection-string
  - dab-config-base64
• Environment variables:
• MSSQL_CONNECTION_STRING=secretref:mssql-connection-string
• DAB_CONFIG_BASE64-secretref:dab-config-base64
• Ingress: External on port 5000
Users report that the /health endpoint returns a healthy response, but queries to an entity named Products fail with a connection error. The SQL login and database existence are confirmed.
You need to ensure the container app can connect to the Azure SQL logical server without changing container app deployment settings or the DAB configuration file.
What should you do on the Azure SQL logical server?
A. Create an auto-failover group for ProductsDB
B. Run DBCC CHECKDB on ProductsDB
C. Create a firewall rule that allows a start and end IP address of 0.0.0.0
D. Enable FORCE_LAST_GOOD_PLAN automatic tuning for ProductsDB
Answer- C

Q-HOTSPOT

You have an azure SQL database that has Query Store Enabled.Query Performance insight shows that one stored procedure has the longest runtime.The procedure run the following parameterized query

```sql
CREATE OR ALTER PROCEDURE dbo.GetRecentOrders
    @CustomerId int,
    @SinceDate datetime2
AS
SELECT TOP (50)
    o.OrderId, o.OrderDate, o.Status, o.TotalAmount
FROM dbo.Orders AS o
WHERE o.CustomerId = @CustomerId
    AND o.OrderDate >= @SinceDate
ORDER BY o.OrderDate DESC;
```

The dbo.Orders table has approximately 120 million rows.
• Customerld is highly selective.
• OrderDate is used for range filtering and sorting.
Existing Indexes
• Clustered index: PK_Orders on (Orderld)
• Nonclustered index: IX_Orders_OrderDate on (OrderDate) with no included columns Execution Plan (slow runs)
• Index seek on IX_Orders_OrderDate
• Key Lookup (Clustered) on PK_Orders for Customerld, Status, TotalAmount
• Sort operator before TOP (50) because results are ordered by OrderDate DESC

Answer Area

Statements	Yes	No
To avoid the explicit sort for the query, create a nonclustered index on `(CustomerId, OrderDate DESC)` that includes `(Status, TotalAmount)`.	✓	○
To eliminate the sort and make the query use an ordered seek, add `CustomerId` as an included column to `IX_Orders_OrderDate`.	○	✓
The plan indicates a bottleneck from a suboptimal query plan, rather locking or blocking.	✓	

Question 57: You have an Azure SQL database named SalesDB on a logical server named sales-sql01. You have an Azure App Service web app named OrderApi. You also have a user-assigned managed identity named OrderApi-Id.
You need to configure OrderApi to connect to SalesDB using Microsoft Entra authentication, granting read and write permissions.
Which Transact-SQL statements should you run in SalesDB?

A. CREATE LOGIN [OrderApi-Id] FROM EXTERNAL PROVIDER;
ALTER ROLE db_datareader ADD MEMBER [OrderApi-Id];
ALTER ROLE db_datawriter ADD MEMBER [OrderApi-Id];

B. CREATE USER [OrderApi-Id] WITH PASSWORD = 'P@ssword!';
ALTER ROLE db_datareader ADD MEMBER [OrderApi-Id];
ALTER ROLE db_datawriter ADD MEMBER [OrderApi-Id];

C. CREATE USER [OrderApi-Id] FROM EXTERNAL PROVIDER.
ALTER ROLE db_datareader ADD MEMBER [OrderApi-Id];
ALTER ROLE db_datawriter ADD MEMBER [OrderApi-Id];

D. CREATE LOGIN [OrderApi-Id] WITH PASSWORD = 'P@ssword!';
ALTER SERVER ROLE sysadmin ADD MEMBER [OrderApi-Id];
Answer-C

Question 58: You have an Azure subscription. The subscription contains an Azure SQL database named SalesDB and an Azure App Service app named sales-api.
• sales-api uses virtual network integration to a subnet named vnet-prod/subnet-app and reads from SalesDB.
• You need to configure authentication and network access to meet the following requirements:
-Ensure that sales-api connects to SalesDB by using passwordless authentication.
-Ensure that all the database traffic remains within the subscription.
• The solution must minimize administrative effort.
What should you configure? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Question 59: You have an Azure SQL database that supports the OLTP workload of an order-processing application.
During a 10-minute incident window, you run a dynamic management view query and discover the following:
• Session 72 is sleeping with open_transaction_count = 1.
· Multiple other sessions show blocking_session_id = 72 in sys.dm_exec_request
• sys.dm_exec_input_buffer(72, NULL) returns only:
-BEGIN TRANSACTION UPDATE Sales.Orders
• Users report that updates to Sales.Orders intermittently time out during the incident window.
• The timeouts stop only after you manually terminate session 72.
What is a possible cause of the blocking?

A. A long-running SELECT statement is blocking writers.
B. Session 72 caused a deadlock.
C. An explicit transaction was started but not committed or rolled back
D. A lock escalation occurred.
Answer- C

Question 60: You have a GitHub Enterprise subscription. Your team is developing an Azure SQL dataset solution from a locally cloned GitHub repository by using Visual Studio Code and GitHub Copilot Chat. A mix of GitHub Copilot instructions is configured at different levels: organization-wide, repository-wide, agent-specific, and personal.
Which instructions will take precedence over the others?
A. repository-wide
B. personal
C. organization-wide
D. agent-specific
Answer- B

Question 61: You have an Azure SQL database that supports a customer-facing API. The API calls a stored procedure named dbo.GetCustomerOrders thousands of times per hour. After a deployment that updated indexes and statistics, users report that the API endpoint backed by dbo.GetCustomerOrders is slower. In Query Store, the same query now has two persisted execution plans. During the last hour, the newer plan had a significantly higher average duration and CPU time than the older plan.
You need to restore the previous performance quickly, without changing the API code. Which Transact-SQL command should you run?
A. EXEC sys.sp_query_store_set_hints
B. DBCC FREEPROCCACHE
C. EXEC sp_query_store_force_plan
D. ALTER DATABASE
Answer- C

Question 62: What is a key benefit of embedding vectors in SQL databases?
A. Faster backups
B. Improved indexing
C. Semantic search capability
D. Reduced storage
Answer- C

Question 63: You have an Azure Al Search service and an index named hotels that includes a vector field named DescriptionVector. You query hotels by using the Search Documents REST API. You need to implement a hybrid search query that uses DescriptionVector and includes captions.
How should you complete the REST request body?
To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all.

Values

- "default"
- "extractive"
- "full"
- "generative"
- "hotels"
- "hybrid"
- "none"
- "semantic"
- "simple"

Answer Area

```
{
 "search": "ocean view",
 "queryType": "semantic" ,
 "semanticConfiguration": "hotels" ,
 "captions": "extractive" ,
 "top": 10,
 "hybridSearch": { "maxTextRecallSize": 50 },
 "vectorQueries": [
    {
      "kind": "vector",
      "vector": [/* embedding array */],
      "fields": "DescriptionVector",
      "k": 50
    }
 ]
}
```

Question 64: You have an SDK-style SQL database project named MyDatabaseProject.sqlproj stored in a private GitHub repository. The repository contains the following GitHub Actions workflow:

Yaml

```yaml
name: Build and Deploy SQL Project

on:
  push:
    branches:
      - main
  workflow_dispatch:

jobs:
  build-and-deploy:
    runs-on: ubuntu-latest
    permissions:
      id-token: write
      contents: read
    steps:
      - uses: actions/checkout@v4
      - name: Build SQL project
        run: dotnet build MyDatabaseProject.sqlproj
      - name: Publish DACPAC
        uses: azure/sql-action@v2
        with:
          action: publish
          path: bin/Debug/MyDatabaseProject.dacpac
          connection-string: ${{ secrets.AZURE_SQL_CONNECTION_STRING }}
```

Additional context:

- The repository contains the AZURE_SQL_CONNECTION_STRING secret.
- The target is an Azure SQL database that allows access to Azure services and is configured to support mixed authentication.
- The workflow runs successfully.

Answer Area

Statements	Yes	No
The workflow relies on a Microsoft Entra workload identity to access the target Azure SQL database.		
Changing the `Build SQL project` step to `run: dotnet build MyDatabaseProject.sqlproj -c Release` will result in a successful deployment.		
The workflow can be triggered manually, without making changes to the repository.		

Question 65: You have an Azure SQL database named SalesDB and an Azure App Service app named sales-api.

- SalesDB contains a table named dbo.Customers with columns CreditCardNumber and Tenantld.
- Currently, sales-api connects using SQL authentication with stored username and password.

You need to recommend a solution that meets the following requirements:

- Provides a passwordless method for sales-api to access SalesDB.
- Ensures that credit card numbers are NOT stored as plain text.

What should you include in the recommendation?

A. Azure Key Vault and Always Encrypted
B. Transparent Data Encryption (TDE) and row-level security RLS)
C. managed identities and dynamic data masking
D. managed identities and Always Encrypted
Answer- D

Question 66: Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution. After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have a SQL database in Microsoft Fabric that contains a table named dbo.Orders

- dbo.Orders has a clustered index.
- Contains three years of data.
- Partitioned by a column named OrderDate by month.

You need to remove all the rows for the oldest month. The solution must queries that access the data in dbo.Orders.
Solution: Run the following Transact-SQL statement:

```sql
Sql

DELETE FROM dbo.Orders
WHERE OrderDate < DATEADD(month, -36, SYSUTCDATETIME());
```

Does this meet the goal?

A. Yes
B. No

Option	Answer	Explanation
A. Yes		Incorrect. A normal `DELETE` statement does not take advantage of table partitioning to efficiently remove an entire partition.
B. No	☑ Correct	The `DELETE` operation removes rows individually and generates transaction log activity. It can acquire locks and increase blocking, which does not minimize the impact on other queries.

Correct Answer:
B. No

- Using DELETE removes rows one by one, generating transaction log entries for each row.
- This is slow and blocking, impacting other queries.
- Since the table is partitioned by month, the correct approach is to use partition switching or TRUNCATE PARTITION to drop the oldest partition efficiently.
- DELETE does not minimize impact.

When working with partitioned tables, use partition switching or truncate partition operations to efficiently remove old data. DELETE is not optimal.

Question 67:

You have a SQL database in Microsoft Fabric that contains a table named dbo.Orders.

- dbo.Orders has a clustered index.
- Contains three years of data.
- Partitioned by a column named OrderDate by month.

You need to remove all the rows for the oldest month. The solution must minimize the impact on other queries that access the data in dbo.Orders.

Solution: Run the following Transact-SQL statement:

```sql
Sql

TRUNCATE TABLE dbo.Orders
WITH (PARTITIONS (partition number));
```

Does this meet the goal?

A. Yes
B. No

Correct Answer:
A. Yes

- TRUNCATE TABLE ... WITH (PARTITIONS ...) efficiently removes all rows in a specific partition.
- It avoids logging row-by-row deletes, minimizing impact on performance and other queries.

Question 68: Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution. After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review
screen.

You have an SDK-style SQL database project stored in a Git repository. The project targets an Azure SQL database.

The CI build fails with unresolved reference errors when the project references system objects.

You need to update the SQL database project to ensure that dotnet build validates successfully by including the correct system objects in the database model for Azure SQL Database.

Solution: Add the Microsoft.SqlServer.Dacpacs.Master NuGet package to Does this meet the goal?

A. Yes

B. No

Answer - No.

Question 69: Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution. After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You have an SDK-style SQL database project stored in a Git repository. The project targets an Azure SQL database.
The CI build fails with unresolved reference errors when the project references system objects.
You need to update the SQL database project to ensure that dotnet build validates successfully by including the correct system objects in the database model for Azure SQL Database.
Solution: Add the Microsoft.Azure.SqlDatabase.Dacpacs NuGet package to the project.
Does this meet the goal?

A. Yes
B. No

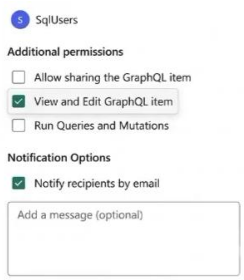
Question 70: Your team is developing an Azure SQL database solution from a locally cloned GitHub repository by using Microsoft Visual Studio Code and GitHub Copilot Chat. You need to ensure that GitHub Copilot Chat uses the team's coding standards when generating Transact-SQL code in Visual Studio Code.
What should you use?
A. .github/copilot-instructions.md
B..vscode/settings.json
C. %APPDATA%\Code\User\settings.json
D. %APPDATA%\Code\User\copilot-instructions.md
Answer-A

Question 71: You are developing an Azure SQL solution by using Microsoft Visual Studio 2026. The solution uses a GitHub repository. You plan to use GitHub Copilot Chat to access the GitHub repository tools by connecting to the GitHub MCP Server. You need to configure Visual Studio to support the planned configuration. The solution must rely on OAuth to access the MCP server.
What should you create?
A. the <SOLUTIONDIR>\.vs\settings.json file that contains the personal access token (PAT) that has the repo scope
B. the <SOLUTIONDIR>\.mcp.json file that contains a URL of https://api.githubcopilot.com/mcp/
C. the <SOLUTIONDIR>\.vscode\settings.json file that contains a URL of https://api.githubcopilot.com/mcp/
D. the <SOLUTIONDIR>\.github\mcp.json file that contains the personal access token (PAT) that has the repo scope
Answer- B

Question 72: Which component handles security in Azure SQL?
A. Azure Key Vault
B. Azure Functions
C. Azure CDN
D. Azure Al Studio
Answer-A

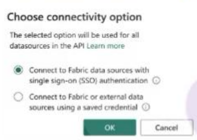
Question 73: Scenario: You have a Microsoft Fabric workspace named Workspacel that contains a SQL

database named SalesDB and an API for GraphQL named SalesApi.You have a Microsoft Entra group named SqlUsers.
From Workspacel, you assign permission to SalesApi as shown in the exhibit:



Statements	Yes	No
The members of SqlUsers can modify the data in SalesDB via SalesAPI.	○	◉
The members of SqlUsers can view the data in SalesDB via SalesAPI.	○	◉
The members of SqlUsers can change the field mappings of SalesAPI.	◉	○

Notification Options:



The connection to SalesDB has the connectivity option configured as:

- **Connect to Fabric data sources with single sign-on (SSO) authentication (selected).**

SqlUsers has the Viewer role for Workspace1.

Question 74: You have an Azure SQL database named DB1 that contains two tables:

- **knowledge_base (contains support articles and embeddings)**
- **query_cache (contains chat questions, responses, and embeddings)**

The database supports an AI-enabled chat agent.
You need to design a solution that meets the following requirements:

- **Serialize the retrieved rows from knowledge_base**
- **Extract the answer field from the response**
- **Extract the embeddings to store in query_cache**

The external large language model (LLM) will be called by using the sp_invoke_external_rest_endpoint stored procedure.

Commands

- AI_GENERATE_CHUNKS
- FOR JSON PATH
- FOR XML PATH
- JSON_QUERY
- JSON_VALUE
- OPENJSON
- VECTOR_DISTANCE

Answer Area

Serialize the retrieved rows from knowledge_base:	FOR JSON PATH
Extract the answer field from the response:	JSON_VALUE
Extract the embeddings to store in query_cache:	JSON_QUERY

Question 75: What is the benefit of using embeddings over keyword search?
A. Smaller storage
B. Exact matches only
C. Context-aware search
D. Faster deletes

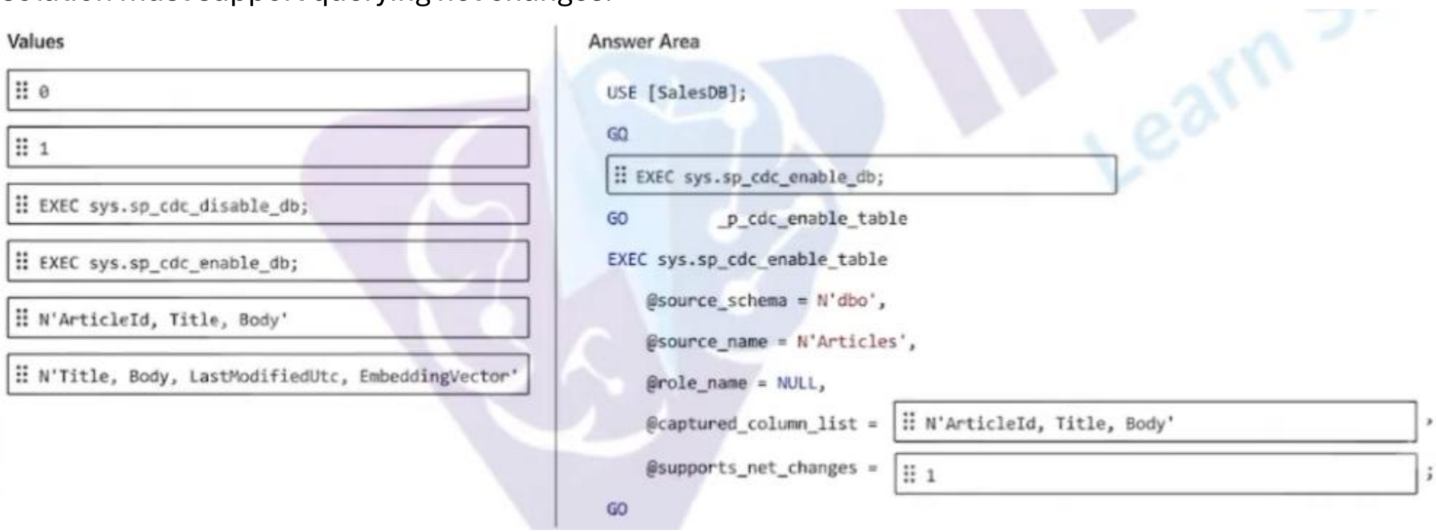
Question 76: You have an Azure SQL database named SalesDB. SalesDB contains a table named dbo.Articles with columns:

- ArticleId
- Title
- Body
- LastModified Utc
- EmbeddingVector

You have an application that generates embeddings from the concatenation Title and Body and stores the results in Embedding Vector.

You plan to implement an incremental embedding maintenance method that will use change data capture (CDC) to update embeddings only for rows that change, without scanning the entire table.

You need to ensure that only the columns required to generate the embeddings are captured. The solution must support querying net changes.



Question 77: Drag and Drop Question You have an Azure SQL database that stores sales data and contains tables named Sales and Products.

- Sales contains three columns: SalesDate, ProductKey, and TotalSale.
- Sales is 10 TB and is loaded nightly by using a batch process.
- Most reporting queries scan large portions of Sales, filter on SalesDate or ProductKey, and use SUM() to aggregate TotalSale.

Products is relatively small and is used primarily for point lookups and joins to Sales.

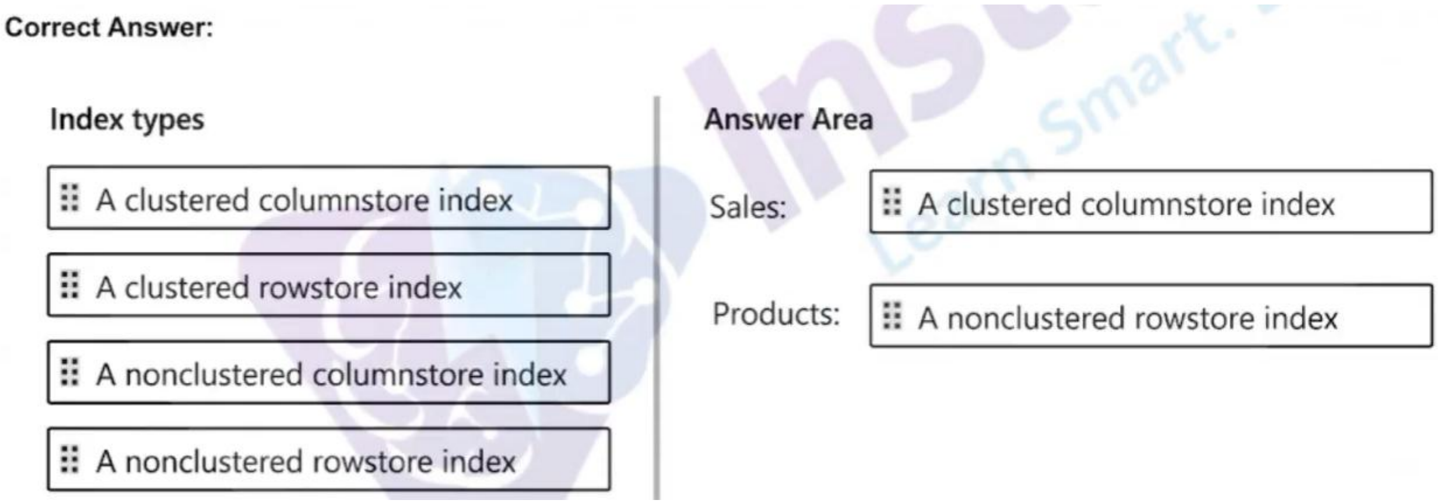
You need to recommend which indexes to create to optimize the reporting queries.

minimize storage requirements.

solution must

Which type of index should you recommend for each table?

To answer, drag the appropriate index types to the correct tables. Each index type may be used once, more than once, or not at all.



Question 78: Which tool helps orchestrate data pipelines for Al workflows?
A. Azure DevOps
B. Azure Data Factory
C. Power BI
D. Azure Monitor
Aswer- B

Question 79: What challenge does Al introduce to SQL systems?
A. Reduced performance always
B. Increased cost only
C. Data privacy and hallucination risks
D. No challenges

Question 80: You have a database named DB1.The schema is stored in a Git repository as an SDK-style SQL database project. You have a GitHub Actions workflow that already runs dotnet build and produces a database artifact.
You need to add a deployment step that publishes the dacpac file to an Azure SQL database by using the secrets stored in GitHub repository secrets. What should you include in the workflow?

A. env:
```
  SQL_CONNECTION_STRING: Server=tcp:myserver.database.windows.net;
...
steps:
- name: Publish
  uses: azure/sql-action@v2
  with:
    action: publish
    path: bin/Debug/db1.dacpac
    connection-string: ${{ env.SQL_CONNECTION_STRING }}
```

B.
```
- name: Publish
  uses: azure/sql-action@v2
  with:
    action: extract
    path: bin/Debug/db1.dacpac
    connection-string: ${{ secrets.SQL_CONNECTION_STRING }}
```

C.
```
- name: Publish
  uses: azure/sql-action@v2
  with:
    action: publish
    path: bin/Debug/db1.dacpac
    connection-string: ${{ secrets.SQL_CONNECTION_STRING }}
```

D.
```
- name: Publish
  run: |
    dotnet build db1.sqlproj \
    /p:TargetConnectionString="${{ secrets.SQL_CONNECTION_STRING }}"
```

Correct Answer:
C. Use azure/sql-action@v2 with action: publish and connection-string: ${{ secrets.SQL_CONNECTION_STRING }}

- action: publish is correct for deploying a dacpac to Azure SQL.
- secrets.SQL_CONNECTION_STRING ensures secure use of GitHub repository secrets.
- Option A incorrectly uses env instead of secrets.
- Option B uses action: extract (wrong direction).
- Option D builds the project but doesn't publish the dacpac.

Question 81: Case Study 2 – Fabrikam

Existing Environment
Fabrikam has a single Azure subscription in the East US 2 Azure region.

The subscription contains an Azure SQL database named DB1.

DB1 contains the following tables:

Employees

Patients

Procedures

UsefulPrompts

ProcedureDocuments

Fabrikam uses Azure Key Vault as a master key.
You have a core premise application named TransientDocsApp that uses a hard-coded username and password in a connection string to access DB1.

Problem Statements
Users report that after executing a long-running stored procedure named sp_UpdateProcedureForPatient, updates to the underlying data are sometimes inconsistent.

Planned Changes
Fabrikam plans to manage all changes to Azure SQL Database objects by using source control in GitHub.

Every pull request submitted to production will be validated before it can be merged.

Requirements
Security requirements:

TransientDocsApp must use Azure Key Vault to manage connection to DB1.

Data stored in DB1 must be encrypted using Transparent Data Encryption (TDE).

Fabrikam must prevent accidental identity of transactions with cryptographic changes to the employee data.

AI requirements:

The UsefulPrompts table in DB1 must NOT be changed by other transactions while the stored procedure runs.

The AI model must be retrained weekly.

The AI model must use Azure Machine Learning Reciprocal Rank Fusion (RRF).

The AI model must use Azure OpenAI embeddings model from Azure OpenAI Service.

The AI model must use Azure OpenAI to generate responses to diagnose patient illness by connecting to an Azure OpenAI endpoint.

Planned Changes
Fabrikam plans to manage all changes to Azure SQL Database objects by using source control in GitHub.

Every pull request submitted to production will be validated before it can be merged.

Requirements
Security requirements:

TransientDocsApp must use Azure Key Vault to manage connection to DB1.

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The UsefulPrompts table in DB1 must NOT be changed by other transactions while the stored procedure runs.

The AI model must be retrained weekly.

The AI model must use Azure Machine Learning Reciprocal Rank Fusion (RRF).

The AI model must use Azure OpenAI embeddings model from Azure OpenAI Service.

The AI model must use Azure OpenAI to generate responses to diagnose patient illness by connecting to an Azure OpenAI endpoint.

Case Study 2

Development requirements:

Fabrikam plans to develop a new REST API named FabrikamAPI that performs the following operations:

Read and insert data into the Transactions table once authenticated.

Execute the dbo.sp_UpdateProcedurePatient stored procedure.

Provide functionality to review a list of the names of patients who underwent medical procedures during the last 30 days.

Table Definition:

Sql

```sql
CREATE TABLE dbo.ProcedureDocuments
(
    DocumentId INT IDENTITY PRIMARY KEY,
    Sourced NVARCHAR(240) NOT NULL,
    Content NVARCHAR(MAX) NOT NULL,
    Embedding VECTOR(768) NOT NULL,
    Created DATETIME NOT NULL DEFAULT SYSUTCDATETIME()
)
```

```json
"entities": {
    "procedures": {
        "source": "dbo.Procedures",
        "rest": true,
        "graphql": true,
        "permissions": [
            {
                "role": "anonymous",
                "actions": [ "read" ]
            }
        ]
    },
    "transactions": {
        "source": "dbo.Transactions",
        "rest": true,
        "graphql": true,
        "permissions": [
            {
                "role": "authenticated",
                "actions": [ "read", "create" ]
            }
        ]
    },
    "UpdateProcedurePatient": {
        "source": "dbo.sp_UpdateProcedurePatient",
        "rest": {
            "enabled": true,
            "method": "post",
            "path": "/procedurepatient"
        },
        "graphql": false,
        "permissions": [
            {
                "role": "authenticated",
                "actions": [ "execute" ]
            }
        ]
    }
}
```

Question 81: You need to use the UsefulPrompts table as defined in the AI requirements. Which stored procedure should you use?

A. sp_addendpoint
B. sp_embeddb
C. sp_AICreate
D. sp_invoke_external_rest_endpoint

Correct Answer:
D. sp_invoke_external_rest_endpoint

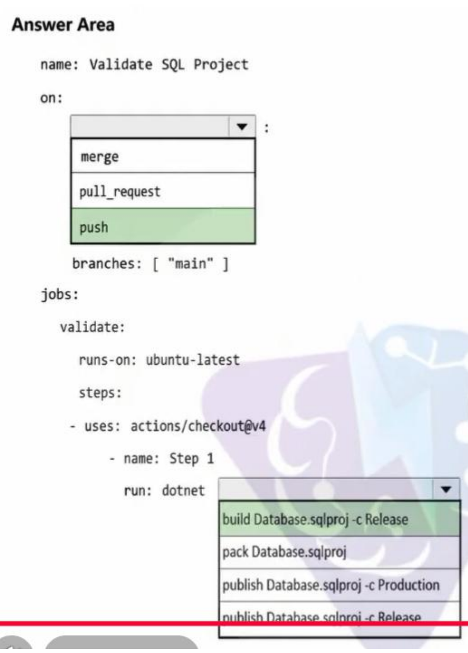
Here's why:

- sp_invoke_external_rest_endpoint is the SQL Server 2025 stored procedure used to call external REST APIs, including Azure OpenAI endpoints.
- This aligns with the AI requirement: "The AI model must use Azure OpenAI to generate responses to diagnose patient illness by connecting to an Azure OpenAI endpoint."
- The UsefulPrompts table provides context for the AI, and the external REST call is how SQL integrates with Azure OpenAI.

Why the others are incorrect:

- A. sp_addendpoint → legacy procedure for Service Broker/HTTP endpoints, not used for AI.
- B. sp_embeddb → not a valid SQL Server procedure.
- C. sp_AICreate → not a valid SQL Server procedure.
- D. sp_invoke_external_rest_endpoint → correct, modern way to call Azure OpenAI securely.

Question 82: You create an SDK-style SQL database project in Microsoft Visual Studio Code named Database.sqlproj and add the project to a GitHub repository.
You need to configure a GitHub Actions workflow to support the planned changes for DB1.
How should you complete the workflow?
To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



Case Study 2

Question 83: Hotspot -

You need to create a solution that meets the development requirements for retrieving the patient transaction data.The solution must ensure that database developers can use the resulting data to join to other tables.

How should you complete the Transact-SQL code? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Case Study 2

Question 84: You need to create a solution that meets the development requirements for retrieving the patient lists.
How should you complete the Transact-SQL code?
To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Correct answers:

Answer Area

```
CREATE PROCEDURE dbo.GetActivePatients
AS
BEGIN
  SET NOCOUNT ON;
 SELECT p.Name
  FROM dbo.Patients AS p
      [dropdown]  (
      WHERE EXISTS
      WHERE NOT EXISTS
      WHERE p.PatienId IN
      WHERE p.PatienId NOT IN
    SELECT PatientId
    FROM dbo.Procedures AS pr
        [dropdown]
        HAVING p.PatientId = pr.PatientId
        HAVING p.TransactionId = pr.TransactionId
        WHERE p.PatientId = pr.PatientId
        WHERE p.TransactionId = pr.TransactionId
      AND pr.TransactionDate >= DATEADD(DAY, -30, SYSUTCDATETIME())
  );
END;
```

1:30:04 / 1:35:30

Case Study 2

Question 85: You need to recommend a solution for the TransactionProcessing application that meets the security requirements.
What should you include in the recommendation?

A. a service principal
B. a system-assigned managed identity
C. a shared access key
D. a user-assigned managed identity

Correct Answer:
A. a service principal

Here's why:

- In the Fabricam case study, the TransactionProcessing application runs outside of Azure.
- Because it is external, it cannot use managed identities (system-assigned or user-assigned), which only work for Azure resources like App Service, VMs, or Functions.
- The secure way to authenticate is via a Microsoft Entra service principal, which can be granted the required roles and permissions.
- Shared access keys are insecure and not recommended.
- Managed identities are correct only when the application runs inside Azure.

Question 86: You have an Azure SQL table that contains the following data:

FaqId	ProductName	Question	Answer
1	Laptop Pro	Does it support USB-C charging?	Yes, it supports USB-C charging.
2	Tablet X	Does it support a stylus?	It supports the Active Pen stylus.

You need to retrieve data to be used as context for a large language model (LLM). The solution must minimize token usage.
Which format should you use to send the data to the LLM?

```
A. [
  {
   "ProductName": "Laptop Pro",
   "Question": "Does it support USB-C charging?",
   "Answer": "Yes, it supports USB-C charging."
  },
  {
   "ProductName": "Tablet X",
   "Question": "Does it support a stylus?",
   "Answer": "It supports the Active Pen stylus."
  }
]
```

```
B. [
  {
   "Prompt": "Yes, it supports USB-C charging."
  },
  {
   "Prompt": "It supports the Active Pen stylus."
  }
]
```

Question 86: You have an Azure SQL table that contains the following data:

FaqId	ProductName	Question	Answer
1	Laptop Pro	Does it support USB-C charging?	Yes, it supports USB-C charging.
2	Tablet X	Does it support a stylus?	It supports the Active Pen stylus.

You need to retrieve data to be used as context for a large language model (LLM). The solution must minimize token usage.
Which format should you use to send the data to the LLM?

Correct answer:

```
A. [
  {
   "ProductName": "Laptop Pro",
   "Question": "Does it support USB-C charging?",
   "Answer": "Yes, it supports USB-C charging."
  },
  {
   "ProductName": "Tablet X",
   "Question": "Does it support a stylus?",
   "Answer": "It supports the Active Pen stylus."
  }
]
```

Here's why:

- Contains only the information needed by the LLM:ProductName
  - Question
  - Answer
- Omits the unnecessary FaqId.
- Provides enough context for the model to associate answers with the correct product.

Best balance between context and token efficiency.

Question 87: You have a SQL database in Microsoft Fabric named SalesBD that contains a table named dbo.Products.
You need to modify SalesBD to meet the following requirements:

- **Create a vector index on the appropriate column.**
- **Use a supplied natural language query vector.**

Correct answers:

Answer Area

```
CREATE VECTOR INDEX idx_products_embedding ON dbo.Products ( [embedding] )
) WITH (METRIC = 'cosine', TYPE = 'DiskANN');
DECLARE @query_vector VECTOR(1536) = @SuppliedVector;
SELECT
   t.product_id,
   t.product_name,
   s.distance
FROM
      [VECTOR_SEARCH] (
      TABLE = dbo.Products AS t,
      COLUMN = embedding,
      SIMILAR_TO = @query_vector,
      METRIC = 'cosine',
      TOP_N = 10
   ) AS s
```

Question 88: You have an Azure SQL database that contains a table named Table1.

- **Table1 contains 25,000,000 rows of data.**
- **It has a datetime2 column named DateKey.**
- **The data in Table1 spans the years 2020 through 2021.**

You need to partition the data in Table1 by year.The solution must minimize how long it takes to rebuild or reindex the table.

Correct answers:

Answer Area

```
CREATE PARTITION FUNCTION PartitionByYear (datetime2)
AS
```

- PARTITION
- RANGE LEFT
- RANGE RIGHT (selected)

```
FOR VALUES (  );
```

- '2019-01-01 00:00:00', '2020-01-01 00:00:00', '2021-12-31 23:59:59'
- '2019-12-31 00:00:00', '2020-12-31 23:59:59'
- '2020-01-01 00:00:00', '2020-02-01 00:00:00', '2020-03-01 00:00:00'
- '2020-01-01 00:00:00', '2021-01-01 00:00:00' (selected)